



Investigation into the Aerodynamic Performance of Multi-Element Airfoils deployed in Horizontal Axis Wind Turbine Blades

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Declaration

I declare that this is my work and that all contributions from other persons have been appropriately identified and acknowledged.

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A handwritten signature in black ink that reads "Aaron Raftery". The signature is written in a cursive style with a long horizontal stroke at the end.

Aaron Raftery

Abstract

This project investigates the effects of incorporating a multi-element airfoil configuration into a Horizontal-Axis Wind Turbine (HAWT) blade. A Siemens 3.6 MW HAWT (the SWT-3.6-120) located at Greater Gabbard offshore wind farm is used for comparison to the proposed multi-element design modification in this project. The HAWT operates in similar environmental conditions to Irish offshore wind turbines and thus gives insight into the capabilities for offshore wind energy off the coast of Ireland. Parameters required for recreating the conditions at Greater Gabbard are available and are used within this project to predict the performance of the multi-element design. Advances in renewable energy technologies, such as the study of multi-element blades, will aid in preventing further environmental damage to the Earth's atmosphere.

The multi-element turbine analysed in this project achieved a maximum power coefficient of approximately 0.476. This is an 8.24% increase in the maximum power coefficient achieved by the baseline HAWT at Greater Gabbard. An angular range of 1° - 27° in the second element's deflection was required to achieve this. A maximum power output of approximately 3.765 MW was achieved by the multi-element turbine at the rated wind speed, with an 8.6% increase from the baseline turbine in energy yield for a four-week period. Although the design is feasible to manufacture, the area of multi-element turbines requires further study before tests on full-scale HAWTs are performed. Due to the cost associated with multi-element blades, this relatively minimal increase in performance may indicate that the technology must be vastly improved before full-scale testing. This project concludes by recommending further investigation into multi-element HAWTs with airfoil configurations comprised of three or more airfoils each of distinct airfoil profile.

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Table of Contents

1. Introduction.....	1
2. Objectives	2
3. Literature Review.....	2
3.1 Climate Change and Energy Usage.....	2
3.2 Wind Energy in Ireland	2
3.3 Ireland's Climate Action Plan	3
3.4 Horizontal-Axis Wind Turbines	3
3.5 Windsail Technology	4
3.6 Multi-Element Airfoils	4
3.7 Baseline Turbine - Greater Gabbard Siemens SWT-3.6-120	5
3.8 Airfoil Choice - The NACA 63-215	7
3.9 Blade Element Momentum Theory and Correction Methods	7
3.10 MSES	8
4. Theoretical Analysis.....	9
4.1 Momentum Theory.....	9
4.2 Blade Element Theory.....	12
4.3 The Lanchester-Betz Limit.....	13
4.4 Blade Element Momentum Theory	13
4.5 Correction Methods.....	14
4.5.1 Prandtl's Approximation for Tip/Root-Loss Factor	14
4.5.2 Snel's Stall Delay Correction.....	15
4.5.3 Viterna Method	16
4.5.4 Modified Glauert Correction.....	17
4.6 Power Output and Energy Yield.....	18
4.7 Chord and Twist Distribution	18
5. Numerical Modelling	19
5.1 MSES	19
5.1.1 Validation of MSES	20
5.2 MATLAB BEMT Code	21
5.2.1 Additional Validation of BEMT Code	21
6. Computational Analysis	22
6.1 Computational Methodology	22
6.2 Computational Setup	23
6.2.1 Operating Conditions	24

6.2.2 Validation of Input Parameters for MATLAB BEMT Code	24
7. Results and Discussion	25
7.1 Predicted Power Coefficient.....	25
7.2 Element Deflection.....	27
7.3 Predicted Power Output	27
7.4 Predicted Energy Yield.....	29
7.5 Chord and Twist Distribution	29
8. Conclusions and Recommendations	30
References.....	I
Appendices.....	A
Appendix A – Study: Multi-Element Airfoil Scale Factor	A
Appendix B – Further Plots for Validation of MSES.....	D
Appendix C – NREL 5 MW Experimental Data.....	E
Appendix D – Further Experimental Conditions for Greater Gabbard.....	G
Appendix E – Additional Data from MSES and BEMT Code.....	H
Appendix E-1 – NACA 63-215 .Airfoil Profile.....	H
Appendix E-2 – Sample MSES Input File.....	I
Appendix E-3 – Sample MSES Plots.....	I
Appendix E-4 – MSES Lift and Drag Coefficients	J
Appendix E-5 - Pitch-Controlled Multi-Element Performance	M
Appendix E-6 – Optimal Twist Distribution.....	M
Appendix F – Further Equations for Prandtl’s Approximation.....	N
Appendix G – MATLAB BEMT Code	O

List of Figures

Figure 1: Airfoils along a blade (Energy.gov 2023).....	1
Figure 2: Inside a HAWT nacelle (Westwood 2012).	1
Figure 3: From left to right; a HAWT, a Gyromill VAWT, and a Darrieus VAWT (Da Rosa 2009).	4
Figure 4: Co-ordinate system for a HAWT, with the x-axis being the axis of rotation (Burton, T. 2011).	4
Figure 5: A possible multi-element airfoil design for a HAWT blade (Ragheb and Selig 2017).	5
Figure 6: TEF oscillation around the flap point (Ruhland and Breitsamter 2021).	5
Figure 7: SWT-3.6-120 wind speed frequency distribution at Greater Gabbard (Argyle <i>et al.</i> 2018).	6
Figure 8: Siemens SWT-3.6-120 experimental data from Greater Gabbard with taken values overlaid (Argyle <i>et al.</i> 2018) (The Wind Power 2018).	6
Figure 9: NACA 63-215 airfoil profile (Airfoil Tools 2024).	7
Figure 10: C_L versus C_D graph taken from a study carried out to validate the accuracy of MSES and XFOIL in predicting the performance of an S406 airfoil ($Re = 5 \times 10^5$) (Maughmer and Coder 2010).	8
Figure 11: Stream tube of a wind turbine (Burton, T. 2011).	9
Figure 12: Pressure and velocity distribution over actuator disc and stream tube (Burton, T. 2011).	9
Figure 13: Particle flowing through the actuator disc with wake rotation (Burton, T. 2011). .	10
Figure 14: Blade element along span of wind turbine blade (Burton, T. 2011).	10
Figure 15: Annular control volume for rotor analysis (Manwell <i>et al.</i> 2010).	11
Figure 16: “Schematic of blade elements; c , airfoil chord length; dr , radial length of element; r , radius; R , rotor radius; Ω , angular velocity of rotor” (Manwell <i>et al.</i> 2010).	12
Figure 17: Blade element along span of a wind turbine blade (Burton, T. 2011).	12
Figure 18: (a) Velocity triangle for a blade element, (b) Force triangle for a blade element (Burton, T. 2011).	12
Figure 19: Lanchester-Betz limit showing the theoretical maximum power coefficient against tip speed ratio as a function of the number of blades, with no drag considered (Manwell <i>et al.</i> 2010).	13
Figure 20: “Span-wise Variation of Combined Tip/root-loss Factor for a Three-blade Turbine Optimized for a Tip Speed Ratio of 6 and with a Blade Root at 20% Span” (Burton, T. 2011).	15
Figure 21: Glauert and Buhl correction methods compared to experimental data and BEMT, with $F = 1$ (Lanzafame and Messina 2007).	17
Figure 22: MSES workflow chart (Drela 2007).	19
Figure 23: Local simulation study and NLR experimental data overlaid pressure distribution for multi-element NLR-7301 with $Re=2.51 \times 10^6$, $M=0.181$, $\alpha=6^\circ$ (Fernandes <i>et al.</i> 2005). ...	20
Figure 24: Procedure for MATLAB BEMT code.	21
Figure 25: Coefficient of power against tip speed ratio performance curve of experimental data from NREL 5 MW wind turbine and predicted performance curve from MATLAB BEMT (parameters used from Table 2) (Jonkman 2014) (NREL 2016).	22
Figure 26: Experimental and numerical comparative approach.	23
Figure 27: Multi-element NACA 63-215 on MSES with Element 2 at 0°	23

Figure 28: Multi-element NACA 63-215 on MSES with Element 2 at 30°.	23
Figure 29: Coefficient of power against tip speed ratio performance curve predicted through BEMT code, with intersecting highlighted maximum mean power coefficient and design tip speed ratio for Greater Gabbard HAWT ($Re = 5 \times 10^6$, $C_{P,max} = 0.44$, Design TSR = 6.56) (Argyle <i>et al.</i> 2018).	25
Figure 30: Accumulated power coefficient versus tip speed ratio curves for the incremental changes in angle of the second element of the multi-element HAWT blade ($Re = 5 \times 10^6$, Design TSR = 6.56).	25
Figure 31: Power coefficient versus tip speed ratio curve showing the highest coefficient of power achieved by varying the second element's angle with wind speed and showing the predicted performance curve of the baseline SWT-3.6-120 HAWT at Greater Gabbard ($Re = 5 \times 10^6$, Design TSR = 6.56).	26
Figure 32: Percentage increase in power coefficient versus tip speed ratio curve showing the increase in power coefficient for the multi-element turbine compared to the baseline SWT-3.6-120 HAWT at Greater Gabbard ($Re = 5 \times 10^6$, Design TSR = 6.56).	26
Figure 33: Second element deflection (°) versus wind speed (m/s) for the multi-element HAWT configuration.	27
Figure 34: Percentage distribution of second element deflection angles for multi-element turbine.	27
Figure 35: Power (MW) versus wind speed (m/s) showing the performance of the baseline project (SWT-3.6-120 at Greater Gabbard) and the multi-element wind turbine (Argyle <i>et al.</i> 2018) (The Wind Power 2018).	28
Figure 36: "Typical power curve of a constant speed stall (dotted) and a variable speed pitch (solid) controlled wind turbine" (Polinder <i>et al.</i> 2003).	28
Figure 37: Chord distribution for multi-element configuration of NACA 63-215 with baseline root and tip chord highlighted to show the feasibility of manufacturing the multi-element wind turbine blade (Siemens 2012).	29
Figure 38: Twist distribution along the blade of the multi-element configuration of NACA 63-215.	30
Figure 39: NACA 63-215 multi-element configurations with scale factors of (0.5, 0.5), (0.75, 0.25), (0.875, 0.125) respectively.	A
Figure 40: Accumulated power coefficient versus tip speed ratio curves for the incremental changes in angle for the second element of the multi-element HAWT blade with a 0.5, 0.5 scale factor for the first and second element respectively ($Re = 5 \times 10^6$, Design TSR = 6.56).	B
Figure 41: Accumulated power coefficient versus tip speed ratio curves for the incremental changes in angle for the second element of the multi-element HAWT blade with a 0.75, 0.25 scale factor for the first and second element respectively ($Re = 5 \times 10^6$, Design TSR = 6.56).	B
Figure 42: Accumulated power coefficient versus tip speed ratio curves for the incremental changes in angle for the second element of the multi-element HAWT blade with a 0.875, 0.125 scale factor for the first and second element respectively ($Re = 5 \times 10^6$, Design TSR = 6.56).	C
Figure 43: Accumulated power coefficient versus tip speed ratio curves for the three scale factor configurations for the elements of the multi-element HAWT blade ($Re = 5 \times 10^6$, Design TSR = 6.56).	C
Figure 44: NLR study. MSES and experimental pressure distribution for multi-element NLR-7301 with $Re=2.51 \times 10^6$, $M=0.181$, $\alpha=6^\circ$ (Fernandes <i>et al.</i> 2005).	D

Figure 45: Local simulation study of MSES using parameters outlined by NLR, with $Re=2.51 \times 10^6$, $M=0.181$, $\alpha=6^\circ$ (Fernandes <i>et al.</i> 2005).....	D
Figure 46: Coefficient of power against tip speed ratio performance curve of experimental data from NREL 5 MW wind turbine (Jonkman 2014) (NREL 2016).....	E
Figure 47: Lift coefficient values of various multi-element airfoils (Ahmed <i>et al.</i> 2013).	G
Figure 48: Reynolds number versus blade length for various MW HAWTs (Ge <i>et al.</i> 2015)...	G
Figure 49: Sample pressure distribution taken from MSES ($Re = 5 \times 10^6$, $\alpha = 0^\circ$, Element 2 deflection = 12°).	I
Figure 50: Intrinsic grid of a NACA 63-215 airfoil at $\alpha = 12^\circ$	J
Figure 51: Estimate for the predicted power output curve of the multi-element design if the power output stabilised at the rated wind speed by adjusting pitch.....	M
Figure 52: Twist angle for various turbines of different airfoils, including distribution of the proposed multi-element wind turbine design in this project to highlight manufacturing feasibility (Taher <i>et al.</i> 2018).....	M

List of Tables

Table 1: Siemens SWT-3.6-120 Datasheet (Siemens 2011).	6
Table 2: Design parameters for NREL 5 MW wind turbine (Jonkman <i>et al.</i> 2009).	22
Table 3: Computational setup for testing using MSES.	23
Table 4: Parameters for multi-element analysis (Siemens 2011) (Ahmed <i>et al.</i> 2013) (Ge <i>et al.</i> 2015).	24
Table 5: Energy yield and power output comparison of baseline turbine to multi-element turbine for 4-week period.	29
Table 6: Chord and twist for multi-element blade design.	29
Table 7: NREL 5 MW Experimental Data (Jonkman 2014) (NREL 2016).	E
Table 8: MSES data used in MATLAB BEMT code.	J

Nomenclature

Symbol	Description	Units
a	Axial induction factor	-
A	Rotor area	m ²
a'	Tangential induction factor	-
A₁	Extrapolation Factor	-
A₂	Extrapolation Factor	-
B₁	Extrapolation Factor	-
B₂	Extrapolation Factor	-
B	Number of turbine blades	-
C_{D,max}	Maximum drag coefficient	-
C_{D,stall}	Stall drag coefficient	-
C_{D,stall}	Drag coefficient at stall	-
C_{D-2D,min}	2-D minimum drag coefficient	-
C_{D-2D}	2-D drag coefficient	-
C_{D-3D}	3-D drag coefficient	-
C_{L-2D}	2-D lift coefficient	-
C_{L-3D}	3-D lift coefficient	-
C_P	Power coefficient	-
C_T	Thrust coefficient	-
ΔC_D	Difference in drag coefficient	-
ΔC_L	Difference in lift coefficient	-
c	Chord length	m
C_D	Drag coefficient	-
C_L	Lift coefficient	-
D	Drag force	N
dr	Radial width	m
F_{hub}	Hub loss factor	-
F_{tip}	Tip loss factor	-
F	Prandtl's combined loss factor	-
h_t	Stagnation enthalpy	J/kg
L	Lift force	N
L/D	Lift-to-drag ratio	-
M	Mach number	-
N	Number of blade elements	-
$\vec{n}$	Euler normal vector	N
P	Power	J/s
p_D⁺	Pressure immediately before rotor disc	Pa
p_D⁻	Pressure immediately after rotor disc	Pa
Q	Torque	Nm
$\vec{q}$	Euler velocity vector	m/s

r	Radial location on the blade	m
R	Blade tip radius	m
Re	Reynold's number	-
T	Thrust	N
U_{∞}	Free stream velocity	m/s
U_D	Net stream-wise velocity	m/s
W	Blade element resultant velocity vector	m/s
λ_r	Local tip speed ratio	-
α_{stall}	Stall angle	° (Degree)
σ'	Local rotor blade solidity	-
α	Angle of attack	° (Degree)
μ	Viscosity	Pa.s
λ	Tip speed ratio	-
Ω	Angular velocity	rad/s
β	Section pitch angle	° (Degree)
ρ	Density	kg/m ³
σ	Blade solidity	-
ω	Angular frequency	rad/s
ϕ	Angle of relative wind	° (Degree)

1. Introduction

In today's world, climate change poses a large threat to humanity's safety. The frequency of storms, floods, droughts, wildfires, and other major climate events is increasing drastically due to human activity (National Geographic 2023). The dependence on fossil fuels to meet global energy demand is slowly being replaced by renewable energies in an effort to reduce greenhouse gas emissions and revert the damage done to the planet. Ireland is at the forefront of this move, with wind energy comprising of approximately 28% of the primary energy produced in the country in 2022 (SEAI 2023). Ireland has the highest percentage energy contribution from onshore wind farms in the world (Wind Energy Ireland 2023). A great interest in wind turbine technology has sparked due to Climate Action Plans set by countries across Europe. Ireland has set carbon emission goals to reach by 2030 with the aim of reaching net zero emissions by 2050 (Government of Ireland 2023).

A wind turbine converts the kinetic energy from the wind into mechanical energy in a shaft that is then converted to electrical energy in a generator (Hansen 2015). This electricity is then transferred into the electrical grid and used to give power to homes and businesses across the country. This process aids towards reaching carbon neutrality. Many jobs are created from this sector as wind energy requires relatively labour-intensive work. However, some say that wind turbines disrupt the views of the natural landscape and that they dislike the noise pollution from the rotating blades. Public feedback is important before planning permission is given in a new region for wind turbines.

Horizontal-axis wind turbines (HAWTs) are the most common type of wind turbine (Manwell *et al.* 2010). Cutting a section along the span of a HAWT blade reveals that the blade section is an airfoil, as illustrated in Figure 1. The pressure difference across each airfoil induced by air flow generates lift and turns the turbine. The power output of the turbine can be predicted using a multitude of methods, the most popular being blade element momentum theory. Two-dimensional lift and drag coefficients are required for calculations within this theory. This data can be obtained by using an aerodynamic analysis software such as MSES, which can analyse single-element and multi-element airfoils under desired environmental conditions.

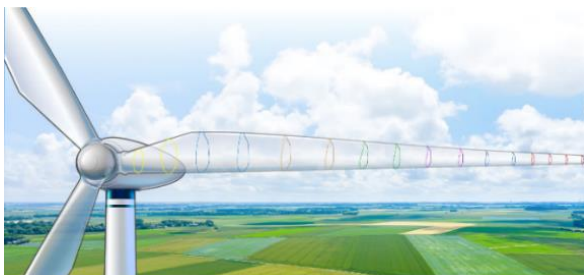


Figure 1: Airfoils along a blade (Energy.gov 2023).

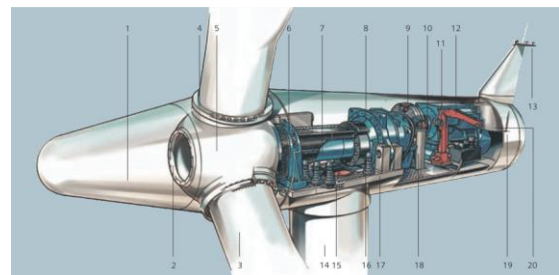


Figure 2: Inside a HAWT nacelle (Westwood 2012).

Multi-element airfoils are common in aviation and sailing, particularly two-element airfoil configurations seen in modern racing yachts. The concept of having a multi-element airfoil design in a full-scale HAWT blade is relatively unexplored. This project investigates incorporating a multi-element airfoil in wind turbine blade design in an effort to increase overall power output.

2. Objectives

The objectives set out for this project are as follows:

- i. Consider a single-element HAWT in similar environmental conditions to Irish wind farms to act as a baseline for comparison with the proposed multi-element design.
- ii. Validate the accuracy of MSES's ability to predict and model aerodynamic forces exerted on a two-dimensional multi-element airfoil by comparison to an external benchmark field test.
- iii. Validate the accuracy of the Blade Element Momentum Theory (BEMT) code's ability to predict and model the performance of a three-dimensional HAWT blade and the operating conditions from the baseline project to be used in the BEMT code.
- iv. Run MSES and the BEMT code to graph coefficient of power versus tip speed ratio curves for the multi-element airfoil configurations under investigation.
- v. Calculate the performance of the multi-element design and determine if there is an improvement in comparison to the baseline single-element HAWT blade at Greater Gabbard.

3. Literature Review

3.1 Climate Change and Energy Usage

Total global energy consumption in 2022 was 604.04 exajoules, of which 45.18 exajoules were generated from renewable energies (EI 2023) (Raftery 2023). In 2022, wind energy capacity increased by 74 GW to a total of 898.824 GW for global wind energy capacity, and a 13.5% worldwide growth in renewable electricity generation. Of all renewable energy, wind contributed the largest percentage to global energy consumption with 2104.8 TWh generated, in comparison to solar generating 1322.6 TWh and all others generating a total of 776.9 TWh. The continuous increase in renewable energy consumption will aid in preventing further climate change.

Growing population has increased energy demands and thus the volume of fossil fuels burned each year (National Geographic 2023). The burning of fossil fuels (e.g., coal, oil, and gas) is a leading cause of drastic global temperature changes (European Commission 2020). Climate change poses an urgent threat to life on Earth, with extreme events of climate disaster becoming more prominent in recent years (The IRC 2023). Catastrophic weather events such as droughts, hurricanes, floods, and heatwaves are destroying wildlife and damaging the Earth's ecosystem.

3.2 Wind Energy in Ireland

Wind energy is Ireland's most used renewable energy source due to the country's geographical position giving powerful and frequent winds both onshore and offshore, with the Irish Sea and Atlantic Ocean providing a huge area for offshore wind farms (Energia 2017). Average annual wind speeds can range from 3-8 m/s across the country, with more frequent high wind speeds coming from the south and west between February and May of each year (Met Éireann 2023). Offshore and onshore wind farms can take this open site wind and convert it to electrical energy to be distributed across Ireland.

Onshore wind farms benefit from the country's highland terrain, with Ireland's mountainous regions providing elevated wind speeds due to the high altitude and lack of obstructions. There are currently over 300 onshore wind farms in the Republic of Ireland, with Galway Wind Park being the highest generating onshore wind farm in the country (Wind Energy Ireland 2023) (SSE Renewables 2017). The wind farm was developed in Connemara's Cloosh Valley in 2017, with 58 Siemens 3 MW wind turbines that are capable of supplying enough wind energy to power over 140,000 Irish homes, offsetting one metric tonne of carbon dioxide emissions for each home in a typical year.

Since 2004, there has only been one operational offshore wind farm off the coast of Ireland (SSE Renewables 2023). Arklow Bank Wind Park consists of seven 3.6 MW wind turbines and is located 10 kilometres off the coast of Arklow, County Wicklow (GE Wind Energy 2003). Submarine cables connect the wind park to ESB Networks by way of the Arklow National Grid Substation and transfer the 25 MW of electrical energy across the country. SSE Airtricity developed the project alongside GE Wind Energy, who supplied and built the wind turbines operated at Arklow Bank. The renewable electricity created is enough to power approximately 16,000 Irish households.

3.3 Ireland's Climate Action Plan

Ireland's Climate Action Plan sets out a roadmap for carbon emission goals and guidelines to reach by 2030, with the aim to reach net zero emissions by 2050 (Government of Ireland 2023). Each sector has outlined goals for the next seven years, with onshore and offshore wind contributing greatly to decreasing the carbon footprint made from electricity generation.

Developmental plans for many offshore wind farms surrounding the country's coast have been made as a part of Ireland's plan to increase wind energy production. The largest of these planned offshore wind farms is the second phase of Arklow Bank Wind Park (SSE Renewables 2023). The project is planned for completion by 2028 and will supply Ireland with a further 800 MW of wind energy. Annually, this wind farm alone will be enough to power 800,000 homes and offset approximately 1 billion kilograms of carbon emissions.

If the guidelines for wind turbine expansion are followed, such as the expansion of Arklow Bank, onshore wind energy is set to reach 6 GW by 2025, and 9 GW by 2030, and offshore wind is expected to reach a minimum of 5 GW by 2030. These guidelines highlight the important role that wind energy will play in Ireland's future.

3.4 Horizontal-Axis Wind Turbines

As discussed previously, the use of wind energy is required to combat the rise in climate change and global disasters. Technological advancements in wind turbine technology are made to match the energy demands of the current times, with the invention of windmills in the 6th century to aid farmers, to the first turbine in the 18th century to generate electricity (Manwell *et al.* 2010). The efficiency of a wind turbine is measured in its ability to extract kinetic energy from oncoming wind. This project focuses on the most commonly used type of wind turbine, the horizontal-axis wind turbine, as opposed to the vertical-axis wind turbine (VAWT) which is far less efficient (Das *et al.* 2017). Both types of turbines are shown in Figure 3 and are

named after their blades' axis of rotation, with the HAWT's axis of rotation being parallel to the ground as shown in Figure 4.

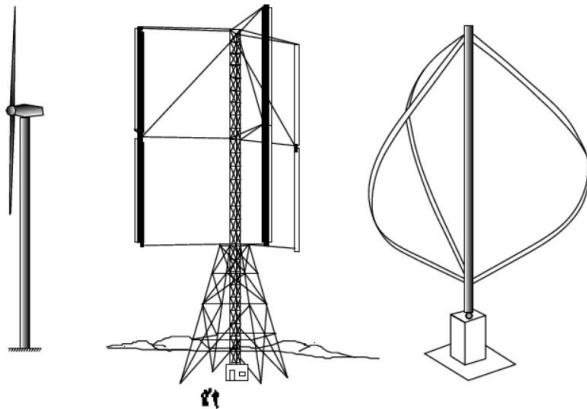


Figure 3: From left to right; a HAWT, a Gyromill VAWT, and a Darrieus VAWT (Da Rosa 2009).

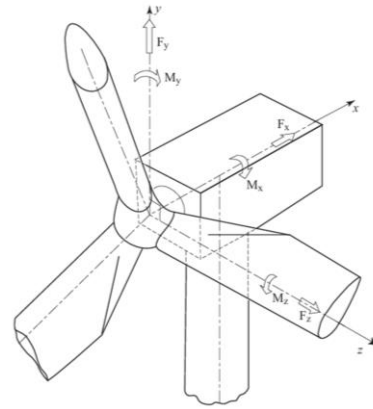


Figure 4: Co-ordinate system for a HAWT, with the x-axis being the axis of rotation (Burton, T. 2011).

HAWT rotor blades consist of airfoils along the span of a blade (Manwell *et al.* 2010). The pressure difference across each of these airfoils generates lift and causes the blades to rotate. HAWTs most commonly have three blades, as increasing past this has minimal impact on performance whilst increasing cost. Other factors influencing modern wind turbine design and prediction methods for power generation are discussed later throughout Section 4, where the methodology behind determining the optimum design of a wind turbine blade to increase performance is detailed further.

3.5 Windsail Technology

Lift-type devices, such as wind turbines, follow the principles behind sailboats sailing cross wind. Modern wing sails, such as those seen in America's Cup, often incorporate multi-element wings to improve aerodynamic performance (Blakeley *et al.* 2012). With this improvement, modern competitive yachts sail upwind at approximately 1.5 times the surrounding wind speed, and downwind at 2-3 times wind speed. To achieve this, most C-Class yachts have a two-element wing arrangement with a 40-60% chord ratio. This configuration keeps the yacht lightweight whilst providing the required lift coefficients and controllability to manoeuvre through the water. This project proposes the idea of implementing a similar multi-element configuration into the design of a HAWT blade and determining the effect this has on the turbine's efficiency.

3.6 Multi-Element Airfoils

Multi-element airfoil configurations are common in nautical and aeronautical design. Minimal research exists on the topic of multi-element blade design in full-scale HAWTs. Introducing the concept of multi-element airfoils into wind turbine blades would improve the blade's aerodynamic characteristics, such as the lift to drag ratio, and ease the development and transportation of larger blades by being comprised of multiple components (Bhavsar *et al.* 2023). Multi-element airfoil design is relatively new in HAWTs and is more commonly used in VAWTs. This is due to the different orientation the flow attacks the VAWT blade (Bah *et al.* 2013). A VAWT multi-element blade design is used to avoid the greater dynamic stall and losses

of kinetic energy to swirl and vortices typically seen during operation. Two-element airfoil configurations have been tested in VAWTs to alleviate these negative effects.

The concept of adjusting the angle of the second element to optimise power output at different wind speeds will be investigated in this project. This is similar to an oscillating trailing edge flap (TEF) used to adjust airfoil camber that is seen in some high-lift devices (Ruhland and Breitsamter 2021). Figure 6 shows an example of an oscillating TEF that was tested to determine the effect on lift and drag coefficients as its angle changed. This test determined through simulation that the average lift and drag increased by 2.1% and 6.6% respectively with the involvement of an oscillating TEF. In this project, the second element will be hinged about the leading edge of the airfoil and will deflect to achieve the maximum power coefficient possible at the oncoming wind speed for the chosen airfoil configuration.

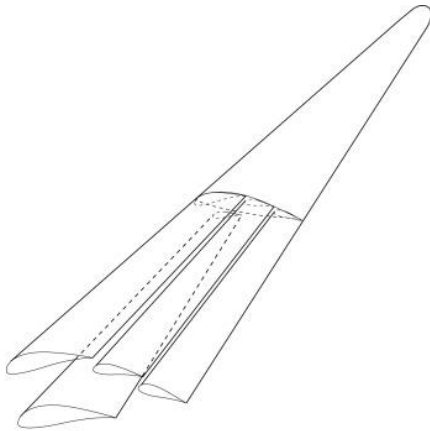


Figure 5: A possible multi-element airfoil design for a HAWT blade (Ragheb and Selig 2017).

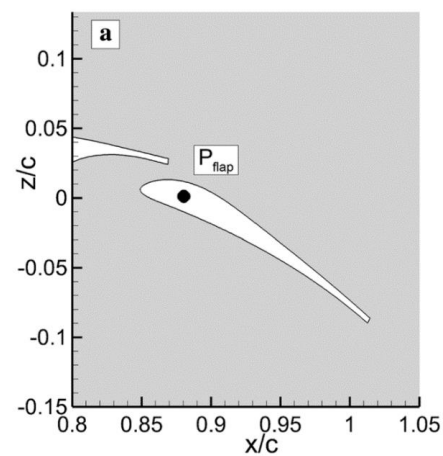


Figure 6: TEF oscillation around the flap point (Ruhland and Breitsamter 2021).

The aerodynamic characteristics of various multi-element configurations will be tested to determine the optimised scale factor and slot spacing of each element. These parameters will be tested based on configurations already investigated in other works for a different airfoil (Kuang *et al.* 2023). Appendix A outlines this study. This project aims to investigate the effect had on aerodynamic performance by implementing a multi-element airfoil configuration in a HAWT blade. Experimental data from a single-element airfoil configuration in a HAWT blade is required to form a baseline for comparison against the multi-element design under consideration. This data will also be used to validate the computational method of determining the performance of the multi-element wind turbine used in this project discussed in Section 3.9.

3.7 Baseline Turbine - Greater Gabbard Siemens SWT-3.6-120

A baseline turbine is required to compare the performance between single and multi-element airfoils for use in wind turbine blades and determine any change in efficiency. The chosen baseline turbine is an SWT-3.6-120 located at the Greater Gabbard Wind Farm off the coast of Suffolk in England. The Greater Gabbard Wind Farm has been operated by SSE Renewables since 2012 (SSE Renewables 2016). The offshore wind farm generates 504 MW of renewable energy from 140 wind turbines. Funding for extension has been granted to potentially double the energy generated at Greater Gabbard. Expansion of offshore wind farms in Ireland is required to meet the goals set out by the Irish Government in the Climate Action Plan. As a

neighbouring country to Ireland, the wind farm at Greater Gabbard gives insight into the capabilities for wind energy off the coast of Ireland.

Parameters, such as diameter, hub radius, and the number of blades of this turbine are available and will be used for further computational setups throughout this project (Siemens 2011). The full list of data required for replicating the environmental and operating conditions at Greater Gabbard is listed in Table 1. Field tests were run on a number of turbines at Greater Gabbard to measure their performance for comparison in a study by Argyle *et al.* (2018). This measured data will be used to compare with the multi-element design configuration. Figure 8 shows the power output versus wind speed for the experimental data taken during the study, with data points overlayed sourced from The Wind Power (2018). Figure 7 shows the wind speed frequency distribution at Greater Gabbard which will be used to calculate the overall energy yield.

Table 1: Siemens SWT-3.6-120 Datasheet (Siemens 2011).

Parameter	Data
Rated Power	3.6 MW
Cut-In Wind Speed	3-5 m/s
Rated Wind Speed	12-13 m/s
Cut-Out Wind Speed	25 m/s
Rotor Diameter	120 m
Swept Area	11,310 m ²
Number of Blades	3
Blade Type	B58
Rotor Tip Speed	82 m/s
Rotor Tilt	6°
Material	GRE
Blade Length	58.5 m
Root Chord	4.2 m
Mean Maximum Power Coefficient	0.44
Aerodynamic Profile	NACA63 Series

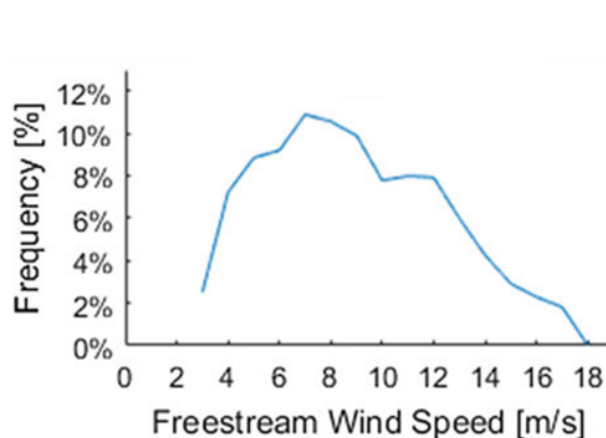


Figure 7: SWT-3.6-120 wind speed frequency distribution at Greater Gabbard (Argyle *et al.* 2018).

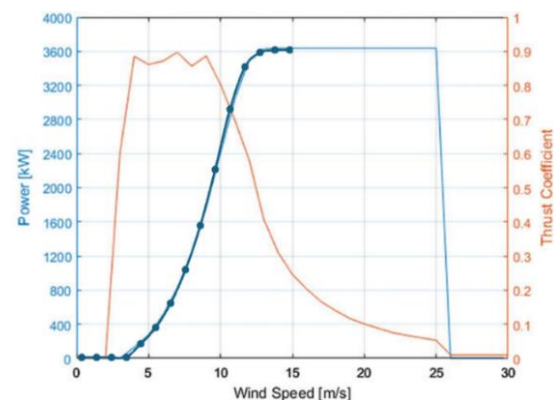


Figure 8: Siemens SWT-3.6-120 experimental data from Greater Gabbard with taken values overlayed (Argyle *et al.* 2018) (The Wind Power 2018).

3.8 Airfoil Choice - The NACA 63-215

An airfoil from the NACA 63-XXX series is used for the B58 blade in the SWT-3.6-120 HAWT. For comparative purposes, an airfoil from this family will be used for investigating the effects of multi-element airfoil design on aerodynamic performance. The NACA 63-215 from the NACA 6-Series was chosen as it has favourable aerodynamic characteristics and is commonly used in the design of HAWTs (Series NF 2012) (Bertagnolio 2001). The 63-215 shows optimal performance at higher speeds, exhibiting a high maximum lift and low drag over the range of operating conditions. Losses in energy production in HAWTs are largely attributed to the maximum achievable lift coefficient along the blade and thus the 6 series is favourable. Airfoils from this series are also commonly used in fighter jets and supersonic jets to take advantage of these characteristics. This project aims to evaluate the effect of incorporating a multi-element design into a pre-existing turbine blade and thus the NACA 63-215 was used for analysis rather than an optimised blade for the given environmental conditions.

The naming convention for the 6-Series is as follows:

- Digit 1: Indicates series of airfoil
- Digit 2: Location of minimum pressure (in tenths of chord)
- Digit 3: The design lift coefficient of the airfoil (in tenths)
- Digits 4 and 5: Thickness (in percentage of chord)

Therefore, the NACA 63-215 is a member of the NACA 6 Series, has a minimum pressure at a position of 0.3 of the chord length, a design coefficient of lift of 0.2, and a 15% thickness as a percentage of chord length.

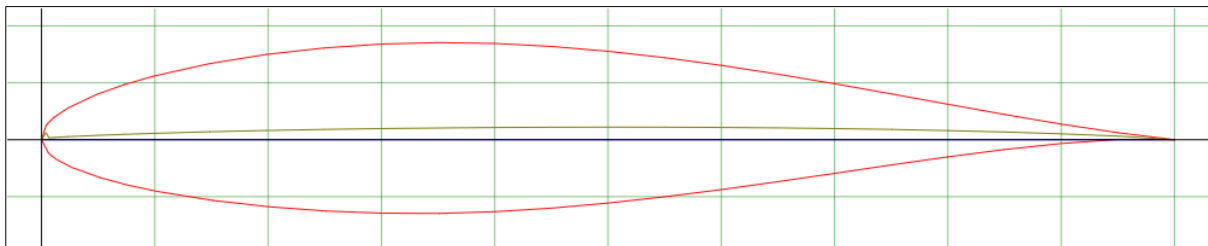


Figure 9: NACA 63-215 airfoil profile (Airfoil Tools 2024).

3.9 Blade Element Momentum Theory and Correction Methods

Blade element momentum theory, otherwise known as strip theory, is a theory used to calculate the performance of a wind turbine rotor (Manwell *et al.* 2010). BEMT is a preferred analysis method for many engineers due to its low computational power and run time in comparison to alternative Computational Fluid Dynamic (CFD) simulation methods. BEMT was originally developed by Betz and Glauert in 1935. This wind turbine analysis method forms the basis of the MATLAB code used in this project. The code was originally developed by Salduskin (2013) and later edited by Foale (2015), and has been used in many projects since. The code outputs a performance curve of the power coefficient against the tip speed ratio of the wind turbine under consideration. This can be used to predict the power output of the turbine at different wind speeds.

BEMT combines momentum theory and blade element theory (Manwell *et al.* 2010). Momentum theory utilises equations from the conservation of linear and angular momentum to analyse the forces over the entire rotor disc, whereas blade element theory analyses the forces

at a cutout section (otherwise known as an element) of the blade. A detailed description of this theory and the correction methods used to improve upon its accuracy in predicting the performance of a wind turbine are discussed later in Section 4. The correction methods used are Prandtl's approximation for tip/root-loss factor, Snel's stall delay correction, the Viterna method, and the modified Glauert correction.

3.10 MSES

Blade element momentum theory requires two-dimensional lift and drag coefficients to calculate the predicted performance of a wind turbine. MSES is a system developed at the Massachusetts Institute of Technology that uses a number of programs to analyse the design of single-element or multi-element airfoils (MIT 2024). This system is suitable for use in this project due to its ability to predict multi-element airfoil behaviour, a feature which many other airfoil analysis programs do not have. A detailed description of the system is discussed later in Section 5, alongside a study performed in this project to validate the system's accuracy in predicting multi-element airfoil aerodynamic behaviour.

Many studies have been conducted outside of this project to validate the accuracy of MSES. Maughmer and Coder (2010) compared a number of theoretical methods of predicting airfoil aerodynamic characteristics. The study concluded that for the theoretical methods analysed, the accuracy of XFOIL and MSES were comparable and that they gave the most accurate prediction in comparison with experimental data. They stated that XFOIL was easier to use and gave better predictions for low wind speeds, but that MSES was necessary for multi-element design and gave better predictions at higher angles of attack. An example from this comparative study is shown below in Figure 10.

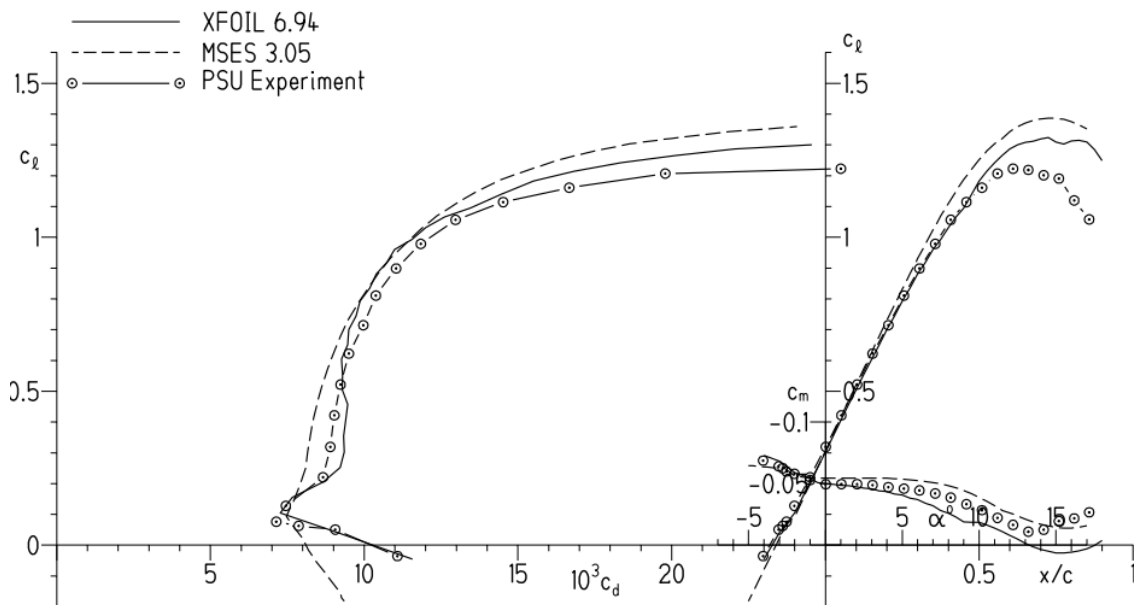


Figure 10: C_L versus C_D graph taken from a study carried out to validate the accuracy of MSES and XFOIL in predicting the performance of an S406 airfoil ($Re = 5 \times 10^5$) (Maughmer and Coder 2010).

4. Theoretical Analysis

The MATLAB code used for this project, originally developed by Salduskin (2013) and later edited by Foale (2015), is based on blade element momentum theory. This theory combines blade element theory and momentum theory to form equations for an iterative coding process. This is a popular method of analysis in comparison to CFD as it is simplified and requires less running time whilst still accurately predicting wind turbine performance (Ledoux *et al.* 2021). This chapter will give explanation to concepts and correction methods used in this BEMT code.

4.1 Momentum Theory

Momentum theory was initially introduced by William J. M. Rankine in 1865, later adapted by Joukowski, Lanchester, and Betz to form the Lanchester-Betz limit, and later again modified by Glauert in 1926 to include the rotating fluids' interaction with the wind turbine blades (see Section 4.5.4) (Ledoux *et al.* 2021). Momentum theory (otherwise known as actuator disk theory) takes a macroscopic approach to analysing the forces on a wind turbine blade. The theory uses the conservation of linear and angular momentum since the force acting on the blades is the rate of change of momentum (Manwell *et al.* 2010). As wind passes through the rotor plane, its velocity decreases and flows into the wake region. This change in velocity is the kinetic energy extracted by the wind turbine. The stream tube illustrated in Figure 11 shows the control volume of air around a wind turbine.

Momentum theory functions under a number of assumptions (Manwell *et al.* 2010):

- The wind operates as a steady, incompressible, homogeneous flow.
- No frictional drag occurs.
- The turbine has an infinite number of rotor blades so can be considered as an actuator disc.
- The thrust over the actuator disc is uniform.
- The wake is non-rotating.
- The static pressure is considered to be undisturbed ambient static pressure far upstream and downstream of the actuator disc, as shown in Figure 12.

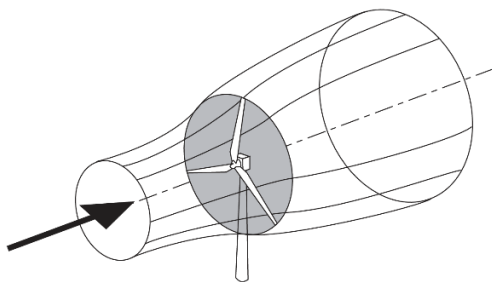


Figure 11: Stream tube of a wind turbine (Burton, T. 2011).

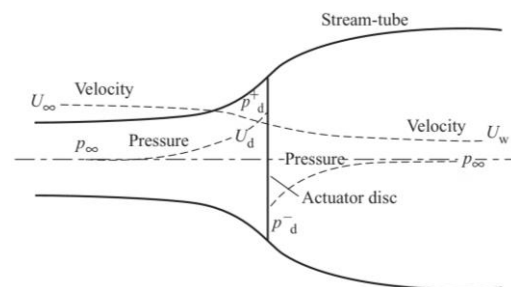


Figure 12: Pressure and velocity distribution over actuator disc and stream tube (Burton, T. 2011).

BEMT is not distinct to a specific type of turbine. The theory considers the flow through a control volume enclosed by a stream tube and its two cross sectional ends (Manwell *et al.* 2010). The turbine itself is solely represented by the actuator disc, and combined with the assumption that there is an infinite number of blades, only the circular area of the disc is used in analysis and equations for flow characteristics are derived. The stream-wise component of the induced velocity variation is represented by $-aU_\infty$, where the axial induction factor, a , is the fractional

decrease in wind velocity as the flow passes through the rotor plane and U_∞ is the free stream velocity far upstream (Burton, T. 2011). The net stream-wise velocity at the actuator disc is:

$$U_D = U_\infty(1 - a) \quad (\text{Equation 1})$$

Or alternatively, the axial induction factor can be given by:

$$a = \frac{U_\infty - U_D}{U_\infty} \quad (\text{Equation 2})$$

The free stream flow is assumed to be steady, incompressible, and homogeneous. Upon passing through the actuator disc, the flow is disturbed and creates a tangential velocity component that opposes the rotation of the wind turbine blades, as shown in Figure 13 (Burton, T. 2011). The torque exerted by the air on the rotor requires the rotor to impose an equal and opposite torque onto the air. As a result, the air gains angular momentum and causes a tangential velocity component in the wake of the disc. Any gain in rotational motion occurs across the disc, as shown in Figure 14, and is expressed in terms of a' , the tangential induction factor.

$$a' = \frac{\omega}{2\Omega} \quad (\text{Equation 3})$$

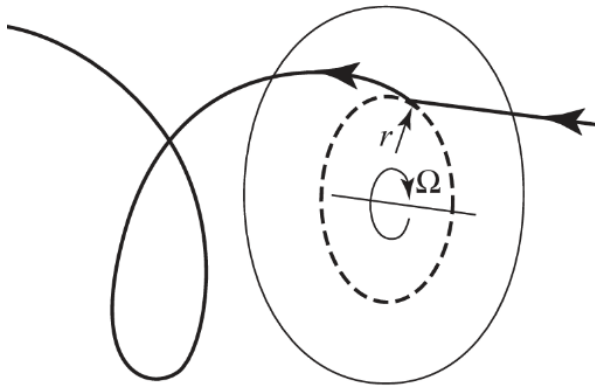


Figure 13: Particle flowing through the actuator disc with wake rotation (Burton, T. 2011).

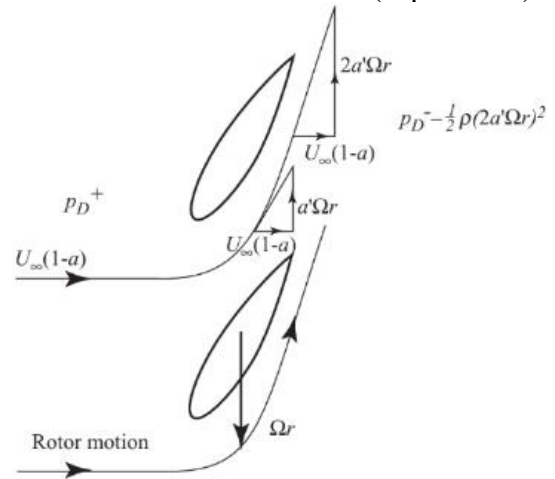


Figure 14: Blade element along span of wind turbine blade (Burton, T. 2011).

The annular stream tube model shown in Figure 15, with notation of radius r and width dr , illustrates the annular elements in which differential thrust and differential torque are exerted (Manwell *et al.* 2010). From the conservation of linear momentum, these concepts are expressed in terms of axial and tangential induction factors in Equations 4 and 5 respectively.

$$dT = 4a(1 - a)\rho U_\infty^2 \pi r dr \quad (\text{Equation 4})$$

$$dQ = 4a'(1 - a)\rho U_\infty \Omega \pi r^3 dr \quad (\text{Equation 5})$$

The tip speed ratio, λ , is defined as the ratio of the blade tip speed to the free stream wind speed (Manwell *et al.* 2010). The local tip speed ratio, λ_r , is the ratio of rotor speed to the wind speed at a desired radius along the span of the wind turbine blade. They are given by:

$$\lambda = \frac{\Omega R}{U_\infty} \quad (\text{Equation 6})$$

$$\lambda_r = \frac{\Omega r}{U_\infty} \quad (\text{Equation 7})$$

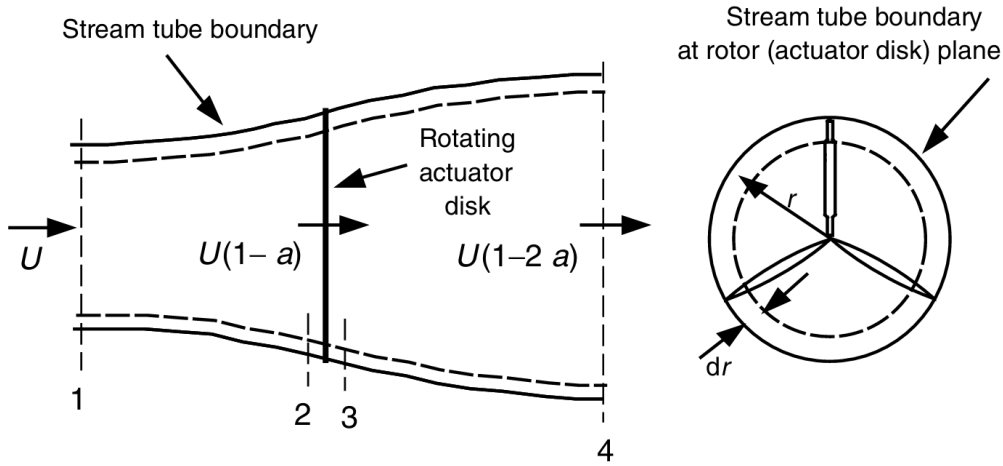


Figure 15: Annular control volume for rotor analysis (Manwell *et al.* 2010).

The power generated at each annular element is expressed by $dP = \Omega dQ$. By substituting dP , λ , and λ_r the power generated can be expressed as:

$$dP = \frac{1}{2} \rho A U_\infty^3 \left[\frac{8}{\lambda^2} a' (1-a) \lambda_r^3 d\lambda_r \right] \quad (\text{Equation 8})$$

C_P is the non-dimensional power coefficient that represents the fraction of power in the wind that the rotor can extract (Manwell *et al.* 2010). From this, C_P can be expressed as:

$$C_P = \frac{8}{\lambda^2} \int_0^\lambda a' (1-a) \lambda_r^3 d\lambda_r \quad (\text{Equation 9})$$

Simply, the power coefficient can be defined as:

$$C_P = \frac{\text{Rotor Power}}{\text{Available Power in the Wind}} \quad (\text{Equation 10})$$

Wind flowing through the actuator disc causes a change in momentum (Burton, T. 2011). Surrounding air is at atmospheric pressure, and thus the pressure difference on the blade ($p_D^+ - p_D^-$) when the wind crosses through the disc causes a thrust force, T , to act on the air:

$$T = (p_D^+ - p_D^-) A_D = 2\rho A_D U_\infty^2 a(1-a) \quad (\text{Equation 11})$$

Thrust is concentrated at the actuator disc and thus the rate of work can be expressed in terms of the thrust and the net stream-wise velocity at the actuator disc. From this the power extracted by the air is given by:

$$\text{Rotor Power} = T U_D = 2\rho A_D U_\infty^3 a(1-a)^2 \quad (\text{Equation 12})$$

Therefore, the power coefficient simplifies to:

$$C_P = \frac{2\rho A_D U_\infty^3 a(1-a)^2}{\frac{1}{2} \rho U_\infty^3 A_D} = 4a(1-a)^2 \quad (\text{Equation 13})$$

4.2 Blade Element Theory

Blade element theory divides the blade into N equal sections, as shown in Figure 16 and Figure 17, and analyses the lift and drag coefficients of each section as a function of angle of attack (Manwell *et al.* 2010). Blade element theory works under a number of assumptions. It is assumed that no aerodynamic interaction occurs between elements, disallowing radial flow, and that the forces on the blade are only influenced by the lift and drag characteristics of the airfoil shape of the blades. It is assumed in blade element theory that the aerodynamic lift and drag forces that act on an individual element along the blade can be analysed in two-dimensional flow as a singular, identical element at the same angle of attack.

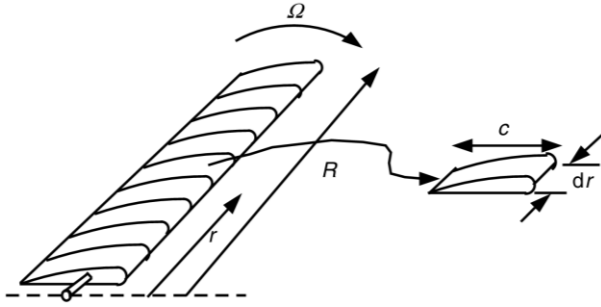


Figure 16: “Schematic of blade elements; c , airfoil chord length; dr , radial length of element; r , radius; R , rotor radius; Ω , angular velocity of rotor” (Manwell *et al.* 2010).

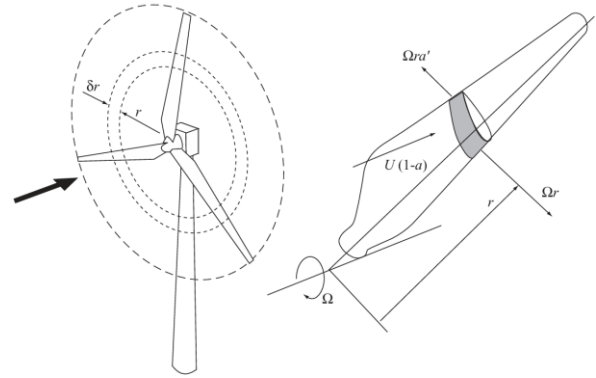


Figure 17: Blade element along span of a wind turbine blade (Burton, T. 2011).

Software, such as XFOIL or MSES, can be used to obtain the lift and drag coefficients (C_L and C_D respectively) for an airfoil at a specific angle of attack. Values for the axial and tangential induction factors can be determined using this two-dimensional data, showing how the aerodynamic characteristics vary with angle of attack (Burton, T. 2011). These values can be estimated through theoretical calculations and refined through an iterative process in blade element momentum theory. The velocity and force triangles on a blade element are shown in Figure 18, of which the magnitude of the resultant relative velocity can be determined using Equation 14.

$$W = \sqrt{U_\infty^2(1-a)^2 + \Omega^2 r^2(1+a')^2} \quad (\text{Equation 14})$$

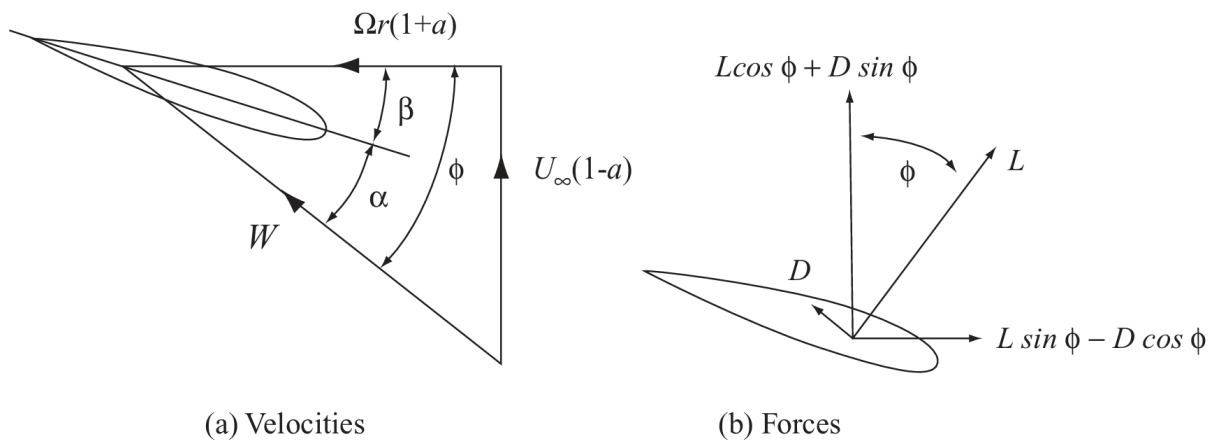


Figure 18: (a) Velocity triangle for a blade element, (b) Force triangle for a blade element (Burton, T. 2011).

The following relationships can be geometrically determined from Figure 18:

$$\alpha = \phi - \beta \quad (\text{Equation 15})$$

$$\tan \phi = \frac{U_\infty(1 - a)}{\Omega r(1 + a)} = \frac{1 - a}{\lambda_r(1 + a')} \quad (\text{Equation 16})$$

4.3 The Lanchester-Betz Limit

The maximum value of C_P can be determined by taking the derivative of the power coefficient (Equation 13) with respect to the axial induction factor (a) and setting it equal to zero (Burton, T. 2011):

$$\frac{dC_P}{da} = 4(1 - a)(1 - 3a) = 0 \quad (\text{Equation 17})$$

Solving this gives the optimal value for the axial induction factor, $a = \frac{1}{3}$, which is used in producing the optimum chord and twist distribution. Substituting back into Equation 13 gives:

$$C_{P,max} = \frac{16}{27} \approx 0.5926 \quad (\text{Equation 18})$$

This is known as the Lanchester-Betz limit and is accredited to Frederic Lanchester (1915), a British aeronautical pioneer, and Albert Betz (1919), a German aerodynamicist. This theoretical maximum efficiency is not due to turbine design, as no design aspects are considered in the formulae. Figure 19 shows the power coefficient approaching this theoretical limit with increasing tip speed ratio for turbines with different numbers of blades.

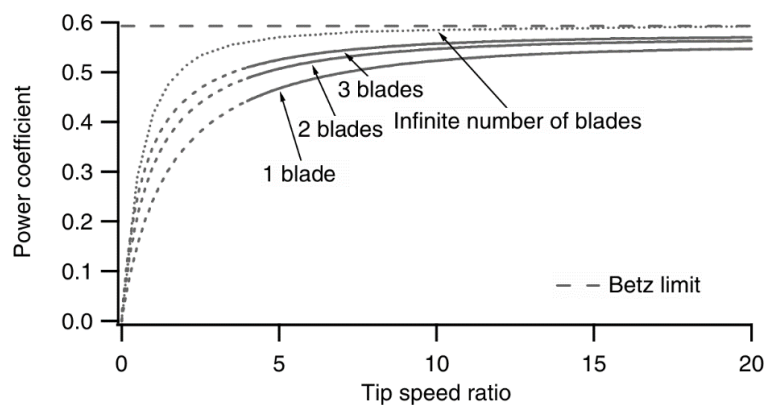


Figure 19: Lanchester-Betz limit showing the theoretical maximum power coefficient against tip speed ratio as a function of the number of blades, with no drag considered (Manwell *et al.* 2010).

4.4 Blade Element Momentum Theory

Blade element momentum theory combines equations from momentum theory and blade element theory. This theory is used to form an iterative process to solve for induction factors and the power coefficient of a turbine at different tip speed ratios. For this theory, it is assumed that the force of a blade element is the sole cause of axial momentum of the air which passes through the actuator disc (Burton, T. 2011). The air is swept by the element and there is no radial interaction between adjacent flows. The iterative process used in the BEMT code created by Salduskin (2013) works to solve the axial and tangential induction factors. Various formulae are required to achieve convergence on the desired tolerance.

To begin the iterative process, an initial estimation for a and a' is required (Manwell *et al.* 2010). The tangential induction factor, a' , is initially estimated to be zero, whereas the axial induction factor is estimated using:

$$a = \frac{1}{4} \left(2 + \pi \lambda_r \sigma' - \sqrt{4 - 4\pi \lambda_r \sigma' + \pi \lambda_r^2 \sigma' (8\beta + \pi \sigma')} \right) \quad (\text{Equation 19})$$

where σ' is the local rotor blade solidity and is defined as:

$$\sigma' = \frac{Bc}{2\pi r} \quad (\text{Equation 20})$$

where B is the number of turbine blades, c is the local chord length of the airfoil, and r is the radius from the centre point of rotation to the blade element.

After algebraic manipulation of Equation 15, a and a' are used in the iterative process by:

$$a = \frac{\sigma' C_L \cos \phi}{1 + 4 \sin^2 \phi} \quad (\text{Equation 21})$$

$$a' = \frac{1}{\left[\frac{4 \cos \phi}{\sigma' C_L} \right] - 1} \quad (\text{Equation 22})$$

where C_L and C_D are the lift and drag coefficients taken from simulation software.

The iterative process required to determine a and a' is as follows (Manwell *et al.* 2010):

- Guess values of a and a' .
- Calculate the angle of the relative wind, ϕ , from Equation 16.
- Calculate the angle of attack, α , from Equation 15 and solve C_L and C_D .
- Update a and a' using Equations 21 and 22.

This process is repeated until an acceptable tolerance for the induction factors has been reached. Upon convergence, the induction factors are inputted into Equation 23 to calculate the power coefficient at different tip speed ratios.

$$C_P = \frac{8}{\lambda^2} \int_{\lambda_h}^{\lambda} \lambda_r^3 a' (1 - a) \left[1 - \left(\frac{C_D}{C_L} \right) \cot \phi \right] d\lambda_r \quad (\text{Equation 23})$$

where λ_h is the local speed ratio at the hub. This method is limited due to its simplicity and thus a number of correction methods are typically implemented to increase accuracy.

4.5 Correction Methods

4.5.1 Prandtl's Approximation for Tip/Root-Loss Factor

Ludwig Prandtl (1919) developed an approximate solution to account for the loss in power production near the tip of the blade (Manwell *et al.* 2010) (Burton, T. 2011). This loss in power is due to the pressure difference at the tip. The suction side of the blade has lower pressure than the pressure side of the blade and causes air to flow around the tip from the lower to upper

surface. This causes a reduction in lift at the tip of the blade, and therefore a reduction in power production. To account for this, a correction factor is introduced into previous equations:

$$F_{tip} = \left(\frac{2}{\pi}\right) \cos^{-1} \left[\exp \left(- \left\{ \frac{\left(\frac{B}{2}\right) \left[1 - \left(\frac{r}{R}\right)\right]}{\left(\frac{r}{R}\right) \sin \phi} \right\} \right) \right] \quad (\text{Equation 24})$$

This is a function of the number of blades, B , the angle of relative wind, ϕ , and the position on the blade, $\frac{r}{R}$. A modified Prandtl tip-loss function is also applied at the blade root to account for zero circulation at the hub.

$$F_{hub} = \left(\frac{2}{\pi}\right) \cos^{-1} \left[\exp \left(- \left\{ \frac{\left(\frac{B}{2}\right) [1 - R_{hub}]}{r \sin \phi} \right\} \right) \right] \quad (\text{Equation 25})$$

The loss factor F is the combination of F_{tip} and F_{hub} . The loss factor along the span of the blade is shown in Figure 20 (Burton, T. 2011). This figure illustrates the power reduction across the blade, with a tip loss factor of one indicating no loss in power reduction. Losses towards the tip and root of the blade are substantial and thus the tip loss factor is introduced into a number of equations to improve accuracy. Appendix F further outlines the equations affected by this loss factor.

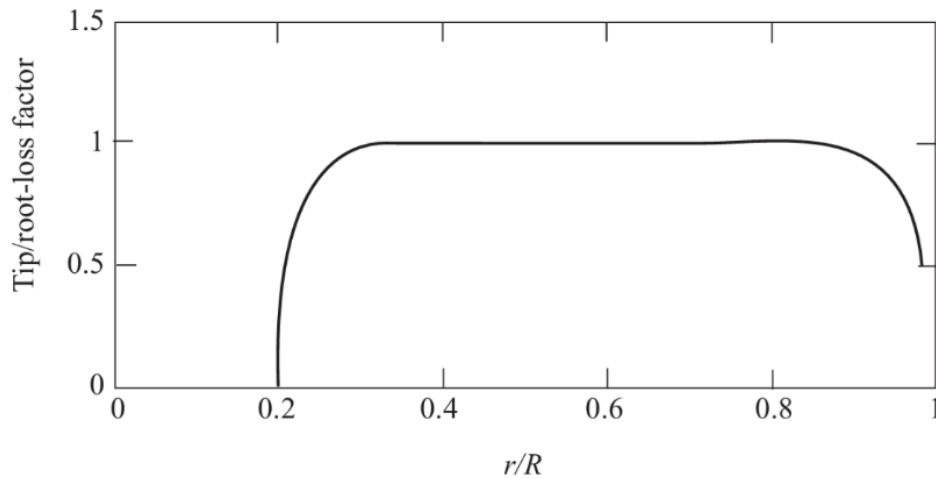


Figure 20: “Span-wise Variation of Combined Tip/root-loss Factor for a Three-blade Turbine Optimized for a Tip Speed Ratio of 6 and with a Blade Root at 20% Span” (Burton, T. 2011).

4.5.2 Snel's Stall Delay Correction

In 1945, a phenomenon was discovered by Himmelskamp when performing tests on propellers (Burton, T. 2011). It was discovered that the lift coefficients taken at the inboard section of a rotating blade were far greater than the maximum possible theoretical lift coefficients for a two-dimensional static test. CFD analyses of rotating blades confirm this decrease in lift coefficient but fail to show what is physically occurring. It is agreed that the blade's rotation causes a decrease in adverse pressure gradient as the flow approaches the trailing edge of the blade. From this, it is also agreed that blade solidity, σ , is the parameter that influences the stall delay. In 1993, Snel *et al.* put forward a modification to standard airfoil two-dimensional, static lift coefficient data that would account for this phenomenon. The modification assumes that if the

linear part of the static, two-dimensional, $C_L - \sigma$ curve is extended beyond stall, let ΔC_L be the difference between the two curves. The correction for the lift coefficient therefore becomes:

$$C_{L-3D} = C_{L-2D} + 3 \left(\frac{C}{r} \right)^2 \Delta C_L \quad (\text{Equation 26})$$

In a similar study, Chaviaropoulos and Hansen (2000) put forward a similar modification to the drag coefficient based on the work of Snel *et al*:

$$C_{D-3D} = C_{D-2D} + a \left(\frac{C}{r} \right)^h \cos^n(\text{twist}) \Delta C_D \quad (\text{Equation 27})$$

where the constants a, h, and n are calibrated using a line of best-fit procedure. The procedure found that for a NACA 63-215; a = 2.2, h = 1, n = 4. ΔC_D is found using:

$$\Delta C_D = C_{D-2D} - C_{D-2D,min} \quad (\text{Equation 28})$$

where C_{D-2D} is the two-dimensional drag coefficient and $C_{D-2D,min}$ is the minimum drag coefficient.

4.5.3 Viterna Method

The Viterna method is used in extrapolating a pre-existing set of airfoil data beyond the stall angle (Mahmuddina 2016). This dataset can come from simulation or experimentation and should contain an initial angle with its associated lift and drag coefficients. The Viterna method is the most commonly used method of extrapolation when using BEMT for its simplicity and reliable results. The forces and moments on each element along the blade are required in BEMT. Model experimentation can only acquire the required data at small angles of attack. The Viterna method assists BEMT to increase the accuracy of the predictive model using two equations:

$$C_L = A_1 \sin 2\alpha + A_2 \frac{\cos^2 \alpha}{\sin \alpha} \quad (\text{Equation 29})$$

$$C_D = B_1 \sin^2 \alpha + B_2 \cos \alpha \quad (\text{Equation 30})$$

where

$$A_1 = \frac{B_1}{2} \quad (\text{Equation 31})$$

$$A_2 = (C_{L,stall} - C_{D,max} \sin \alpha_{stall} \cos \alpha_{stall}) \quad (\text{Equation 32})$$

$$B_1 = C_{D,max} \approx 1.11 + 0.018AR \quad (\text{Equation 33})$$

$$B_2 = \frac{C_{D,stall} - C_{D,max} \sin^2 \alpha_{stall}}{\cos \alpha_{stall}} \quad (\text{Equation 34})$$

The value of AR, the aspect ratio, can be obtained through BEMT but is generally taken between 9-10 for most computations (Mahmuddina 2016).

4.5.4 Modified Glauert Correction

The basic assumptions of BEMT become invalid when the induction factors are greater than approximately 0.4 (Moriarty and Hansen 2005). This is due to flow in the far wake propagating upstream and causing more flow from outside of the wake to enter and increase turbulence. This increases the thrust on the rotor disc whilst the upstream flow slows. As described earlier, Glauert developed a correction method in 1926 to account for this. Buhl (2004) later derived a modification to Glauert's correction method to include Prandtl's tip/root-loss correction factor:

$$C_T = \frac{8}{9} + \left(4F - \frac{40}{9}\right)a + \left(\frac{50}{9} - 4F\right)a^2 \quad (\text{Equation 35})$$

This modification was required as when the losses at the tip increase, the induced velocities also increase and therefore increase the possibility of a turbulent wake near the blade tips. C_T can be used in a new equation to solve for the axial induction factor that incorporates the work of Prandtl, Glauert, and Buhl. This relationship is different from those derived in Manwell (2010) and Burton (2011):

$$a = \frac{18F - 20 - 3\sqrt{C_T(50 - 36F) + 12F(3F - 4)}}{36F - 50} \quad (\text{Equation 36})$$

As stated before, BEMT becomes invalid when the induction factors are greater than approximately 0.4. This is illustrated in Figure 21 and shows that BEMT and the Modified Glauert Correction produce a value of 0.96 for the thrust coefficient. The modified equations in this section are implemented into the MATLAB BEMT code used in this project to increase its accuracy in predicting experimental results.

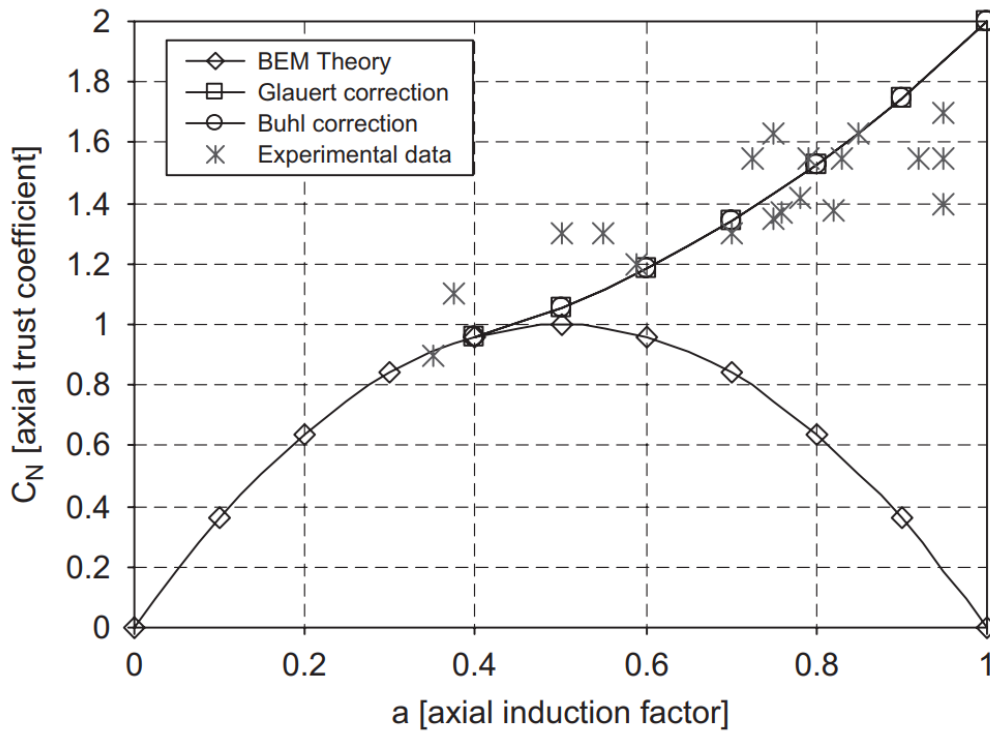


Figure 21: Glauert and Buhl correction methods compared to experimental data and BEMT, with $F = 1$ (Lanzafame and Messina 2007).

4.6 Power Output and Energy Yield

The theory described thus far is used to determine the power coefficient of a turbine at varying wind speed and therefore indicate the performance of the turbine. The standard presentation method for performance is through a coefficient of power against tip speed ratio ($C_p - \lambda$) curve (Burton, T. 2011). The MATLAB BEMT code used for this project outputs this performance curve upon completing the iterative process and incorporating the correction factors previously described. The power coefficient is used in determining the power output of a turbine by combining Equations 10 and 13 to form Equation 37. The power output is then used to calculate the energy yield of the turbine in Equation 38. In addition to graphing the power coefficient against tip speed ratio, the BEMT code used in this project outputs graphs of optimum chord and twist angle along the span of the blade.

$$\text{Total Power Output } (W) = \sum \frac{1}{2} \rho U_{\infty}^3 A_D C_P \quad (\text{Equation 37})$$

$$\text{Total Energy Yield } (Wh) = \sum \text{Power} \times \text{Time} \quad (\text{Equation 38})$$

4.7 Chord and Twist Distribution

A variable speed wind turbine can maintain a constant tip speed ratio to achieve maximum power coefficient despite changes in wind speed (Burton, T. 2011). Optimum blade geometry is needed to achieve the chord and twist distribution required of a variable speed wind turbine blade. To align with the Lanchester-Betz limit discussed earlier, the induction factors for optimised blade operation are:

$$a = \frac{1}{3} \quad (\text{Equation 39})$$

$$a' = \frac{a(1-a)}{\lambda^2 \mu^2} \quad (\text{Equation 40})$$

Fundamental equations from BEMT combine to give equations for the optimum chord and twist distribution of a turbine blade (Burton, T. 2011). Inputting the optimised induction factors as discussed before gives Equation 41, the chord distribution for optimum blade geometry, and Equation 42, the twist distribution for optimum blade geometry:

$$c = \frac{2\pi R}{B\lambda C_L} \left(\frac{\frac{8}{9}}{\sqrt{\left(1 - \frac{1}{3}\right)^2 + \lambda^2 \mu^2 \left(1 + \frac{2}{9\lambda^2 \mu^2}\right)^2}} \right) \quad (\text{Equation 41})$$

$$\tan \phi = \frac{1 - \frac{1}{3}}{\lambda \mu \left(1 + \frac{2}{9\lambda^2 \mu^2}\right)} \quad (\text{Equation 42})$$

where $\mu = \frac{r}{R}$.

5. Numerical Modelling

5.1 MSES

MSES is a system developed at the Massachusetts Institute of Technology (MIT 2024). The system uses a number of programs to analyse the design of airfoils, the relationships of which are illustrated in the flowchart shown in Figure 22 (Drela 2007). The system is used to investigate the effects of changing environmental conditions and geometry of single-element or multi-element airfoils. This system is suitable for this project due to its ability to predict multi-element airfoil behaviour, a feature which many other airfoil analysis programs (such as XFOIL) do not have. The prediction of airfoil performance is based on the steady state conservative Euler equations; the conservation of mass, conservation of momentum, and conservation of energy (Silva *et al.* 2005):

$$\oint \bar{\rho} \vec{q} \cdot \vec{n} ds = 0 \quad (\text{Equation 43})$$

$$\oint (\rho(\vec{q} \cdot \vec{n}) \vec{q}) ds = 0 \quad (\text{Equation 44})$$

$$\oint \bar{\rho} \vec{q} \cdot \vec{n} h_t ds = 0 \quad (\text{Equation 45})$$

where ρ is the fluid density, $\vec{q}$ is the velocity vector, $\vec{n}$ is the normal vector, p is the pressure and h_t the stagnation enthalpy.

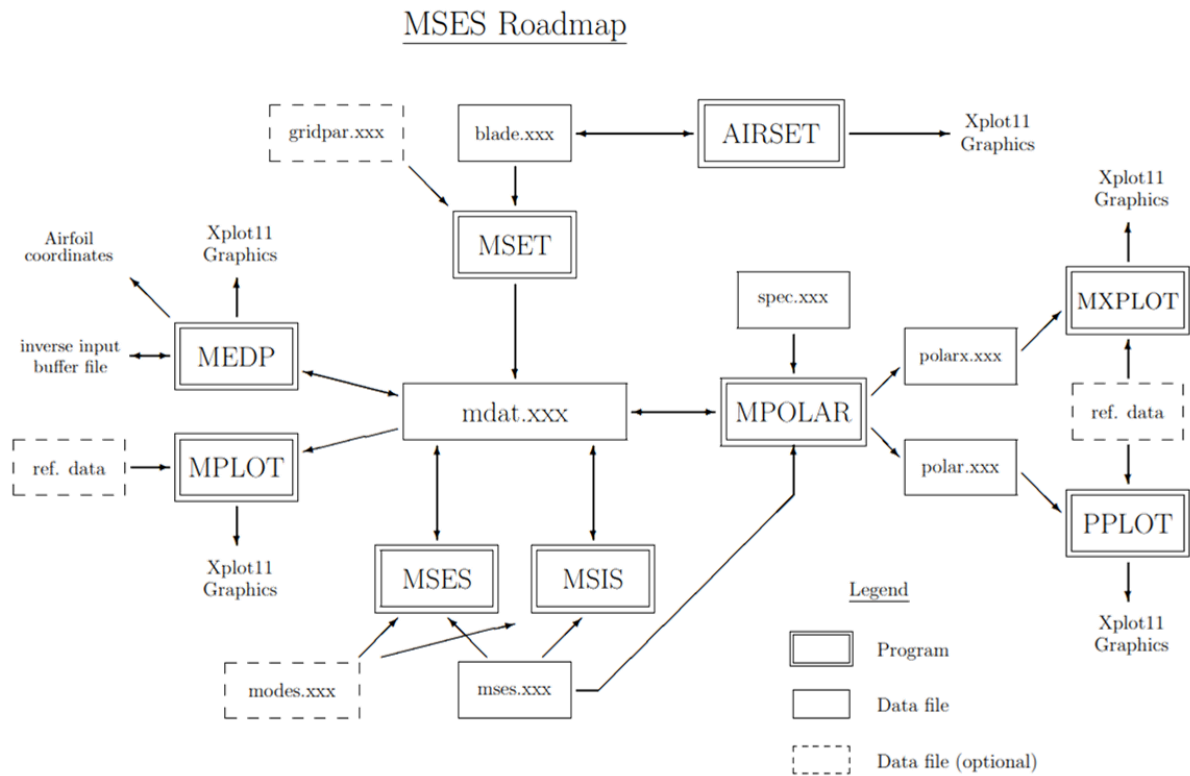


Figure 22: MSES workflow chart (Drela 2007).

The system uses the Euler equations to describe the inviscid part of the flow (Silva *et al.* 2005). Assumptions in the system are based off of Prandtl's boundary layer equations that describe the viscous part of the flow to be restricted by a thin boundary layer, where any viscous effects outside of this are considered negligible. The semi-empirical e^n envelope method was used in the development of MSES (Drela 2007). Drela derived equations to predict boundary layer transition. These equations are discretized in an intrinsic grid, as shown in Figure 50, where each coordinate line represents a streamline around the airfoil's body. The Newton method is then used to simultaneously solve these equations and obtain converged solutions for the two-dimensional lift and drag data required for BEMT simulations. An example pressure distribution curve for a NACA 63-215 multi-element airfoil configuration and an example intrinsic grid for a NACA 63-215 single-element airfoil taken from MSES is shown in Appendix E-3.

5.1.1 Validation of MSES

A wind tunnel analysis of a multi-element NLR-7301 airfoil was performed by the Netherlands Aerospace Centre (NLR) to validate the use of MSES in accurately predicting multi-element airfoil behaviour (Fernandes *et al.* 2005). The pressure distribution across the length of the blade was recorded and compared with simulated data taken from MSES. The experimental data from this study is presented in Appendix B. To validate the use of MSES in this project, the loading conditions from this study were used to run a simulation study. The results of this are shown in Appendix B. Figure 23 below shows the MSES study carried out in this project with experimental data points overlayed to highlight the accuracy of MSES for use in this project. Appendix B shows the individual plots used to create the figure below.

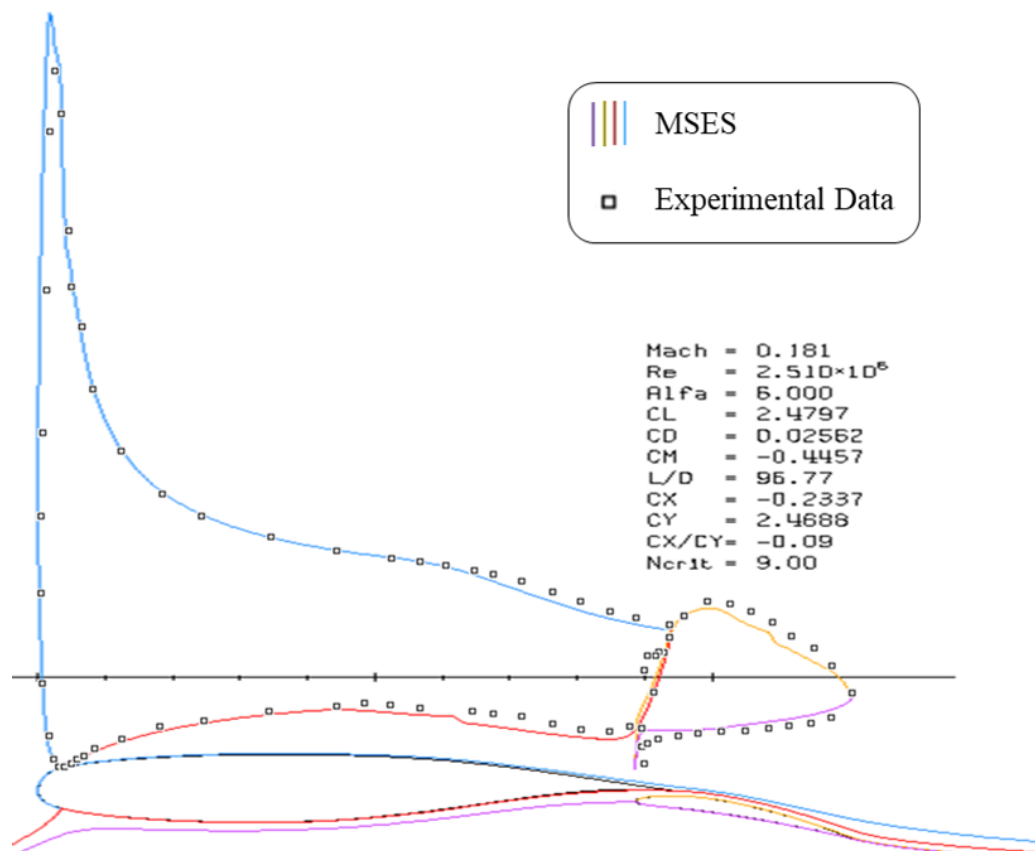


Figure 23: Local simulation study and NLR experimental data overlayed pressure distribution for multi-element NLR-7301 with $Re=2.51 \times 10^6$, $M=0.181$, $\alpha=6^\circ$ (Fernandes *et al.* 2005).

5.2 MATLAB BEMT Code

The MATLAB BEMT code originally developed by Salduskin (2013) uses equations described throughout Section 4. It does this based off of two-dimensional airfoil data taken from MSES and input parameters described later in Section 6.2.1. The code outputs three graphs; coefficient of power versus tip speed ratio performance curve, optimum chord distribution, and optimum twist angle distribution. The code's iterative procedure is detailed in Figure 24, and a copy of the MATLAB script used in this project is in Appendix G.

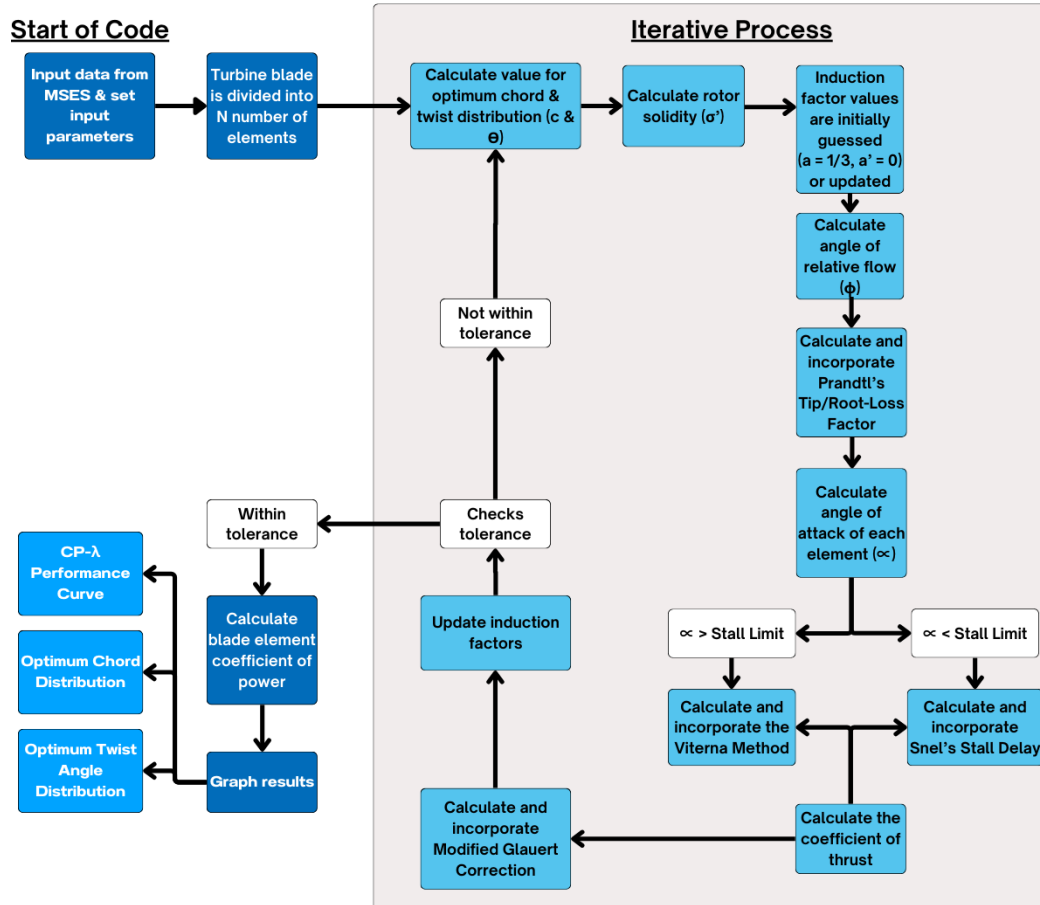


Figure 24: Procedure for MATLAB BEMT code.

5.2.1 Additional Validation of BEMT Code

A study was conducted to determine the accuracy of the MATLAB BEMT code created by Salduskin (2013) in predicting the aerodynamic behaviour of a HAWT. The National Renewable Energy Laboratory (NREL) created a report for a 5 MW offshore wind turbine with suitable data to be used as a baseline for research teams to compare their turbine's performance to (Jonkman *et al.* 2009). The report outlines values for the lift and drag coefficients of the turbine at different angles of attack to be used for comparison in computation. This data was inputted into the MATLAB code of this project, alongside experimental conditions outlined in Table 2, to output the expected performance curve. This was then compared to the freely available power coefficient versus tip speed ratio data made available by the NREL, as shown in Figure 25 (Jonkman 2014) (NREL 2016). It is evident from this graph that the MATLAB BEMT code is sufficiently accurate in predicting the performance of the HAWT analysed by

the NREL. The full list of data, the input file used and further information relating to this study is available in Appendix C. The MATLAB BEMT code is further validated in Section 6.2.2.

Table 2: Design parameters for NREL 5 MW wind turbine (Jonkman *et al.* 2009).

Design Parameter	Value
Rated Wind Speed	11.4 m/s
Rotor Diameter	126 m
Number of Blades	3
Rotor Tip Speed	80 m/s
Blade Length	61.5 m
Freestream Angle of Attack	6.3°

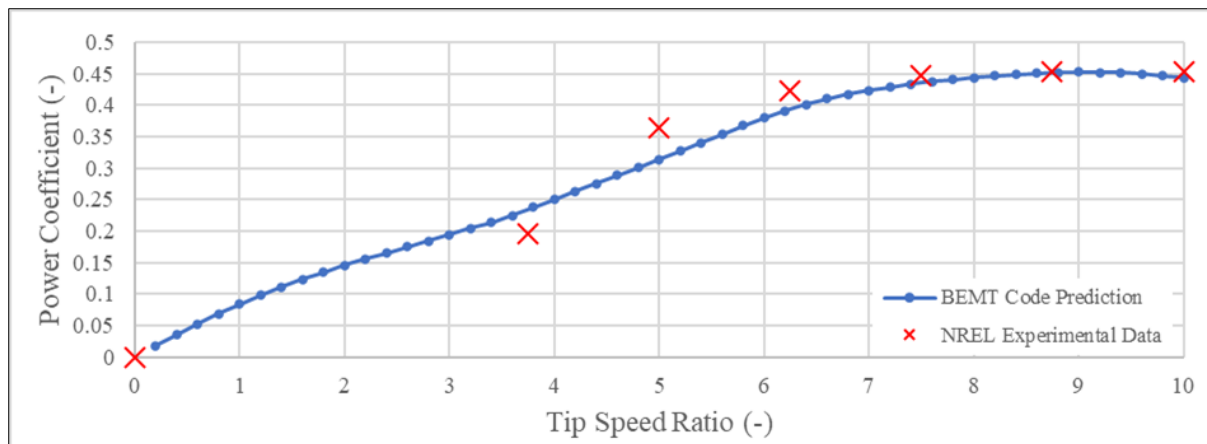


Figure 25: Coefficient of power against tip speed ratio performance curve of experimental data from NREL 5 MW wind turbine and predicted performance curve from MATLAB BEMT (parameters used from Table 2) (Jonkman 2014) (NREL 2016).

6. Computational Analysis

6.1 Computational Methodology

The procedure followed in this project to analyse the aerodynamic performance of the multi-element airfoil blade is outlined below. All software used is validated in Sections 5 and 6.2.2.

1. The chosen multi-element airfoil is setup (as described later in Section 6.2) in MSES. The two-dimensional lift and drag coefficients are recorded at various angles of attack and stored in a text file.
2. The angle of the second element is increased, and step 1 is repeated. This process repeats until the data is recorded for all desired airfoil configurations.
3. The text files are loaded into the MATLAB BEMT code. The code uses this data (as described in Section 5.2) to calculate the performance of the blade at the inputted multi-element configuration.
4. The coefficient of power against tip speed ratio performance curve for each multi-element configuration is recorded and plotted over one another.
5. The optimum power coefficient is determined at different wind speeds and used to estimate the power output and energy yield of the multi-element wind turbine.
6. The results are compared with the performance of the chosen baseline project.

This project aims to numerically predict the performance of a multi-element horizontal-axis wind turbine at the same experimental conditions as the chosen baseline project. This numerical prediction is then compared to the field test experimental performance of the baseline turbine. This approach is common when investigating design alterations to traditional wind turbine blade design. This process is illustrated in Figure 26.

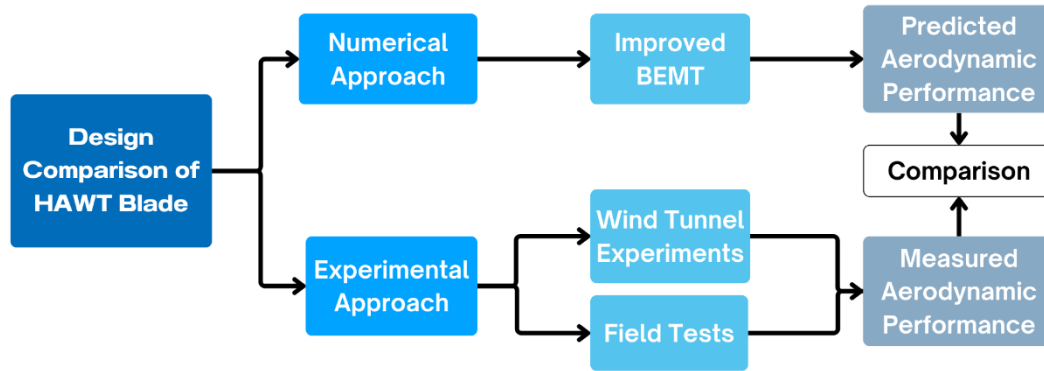


Figure 26: Experimental and numerical comparative approach.

6.2 Computational Setup

The multi-element airfoil chosen for testing in MSES consists of two elements, both of which are of the same airfoil shape. Element 2 deflects in 1° increments from 0° to 30° . Further increments led to stall and were deemed unnecessary for investigation. The slot spacing (the gap and overlap) and scale factor (percentage of chord length) of the two elements remains constant throughout testing and is based on testing shown in Appendix A. The testing performed is based off of other works (Kuang *et al.* 2023). This spacing is outlined in Table 3.

Table 3: Computational setup for testing using MSES.

Airfoil Name	Element 1 Scale Factor	Element 2 Scale Factor	Element 2 Deflection	X-Direction Slot Spacing	Y-Direction Slot Spacing
NACA63-215	0.75%	0.25%	$0^\circ - 30^\circ$	1.8%	1.8%

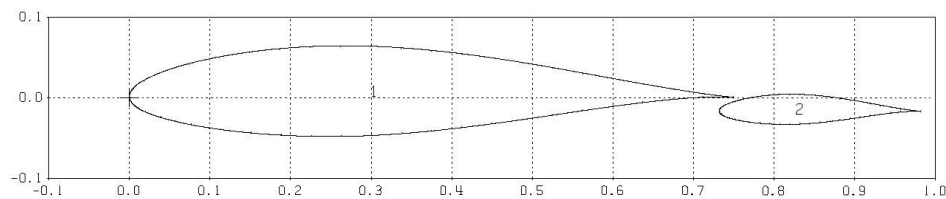


Figure 27: Multi-element NACA 63-215 on MSES with Element 2 at 0° .

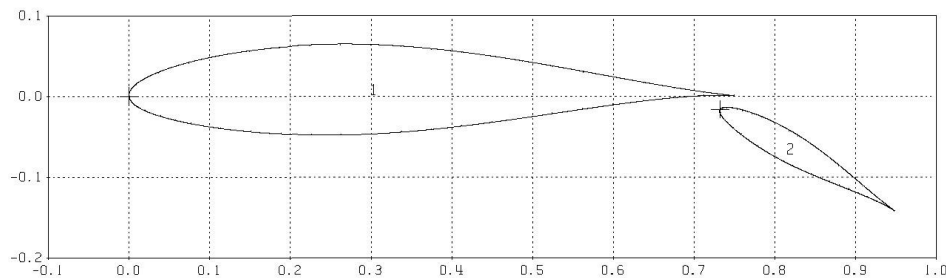


Figure 28: Multi-element NACA 63-215 on MSES with Element 2 at 30° .

6.2.1 Operating Conditions

To replicate the experimental and operating conditions at Greater Gabbard and to ensure a valid performance comparison, a number of values are required to describe the flow conditions used in MSES, as well as the input parameters required for the MATLAB BEMT code. The design lift coefficient for a NACA series multi-element airfoil varies with element layout (Ahmed *et al.* 2013). As shown in Figure 47 in Appendix D, a plain airfoil flap configuration has a design lift coefficient of approximately 2.4. The design tip speed ratio is found by dividing the rated tip speed (82 m/s) by the rated wind speed (12.5 m/s) for the turbine (Siemens 2011). This equates to a design TSR of 6.56. The air density at the turbine's height is 1.225 kg/m³. This is used in calculating the turbine's power output. The freestream angle of attack for use in MSES is taken as the pitch of the airfoil (6°). A freestream Mach number of 0.04 is found by dividing the local flow velocity (12.5 m/s) by the speed of sound in the medium (speed of sound in air is 346 m/s). An approximate freestream Reynolds number of 5×10^6 is found for the 3.6 MW HAWT using Figure 48 in Appendix D (Ge *et al.* 2015). The full list of values used to recreate the experimental conditions of Greater Gabbard are shown in Table 4.

Table 4: Parameters for multi-element analysis (Siemens 2011) (Ahmed *et al.* 2013) (Ge *et al.* 2015).

Design Parameter	Value
Rated Power	3.6 MW
Rated Wind Speed	12-13 m/s
Rotor Diameter	120 m
Swept Area	11,310 m ²
Number of Blades	3
Rotor Tip Speed	82 m/s
Blade Length	58.5 m
Mean Maximum Power Coefficient	0.44
Aerodynamic Profile	NACA63 Series
Air Density	1.225 kg/m ³
Design Lift Coefficient (C _L)	2.4
Design Tip Speed Ratio	6.56
Freestream Angle of Attack	6°
Freestream Mach Number	0.04
Freestream Reynolds Number	5×10^6

6.2.2 Validation of Input Parameters for MATLAB BEMT Code

Freely available data for the Greater Gabbard Siemens SWT-3.6-120, as outlined in Table 4, was used in validating the MATLAB BEMT code and the use of the input parameters in the code. The code outputs the predicted coefficient of power versus tip speed ratio curve as shown in Figure 29. Although the full power coefficient performance curve for the Greater Gabbard turbine is not available, the maximum mean power coefficient and design tip speed ratio is known (Argyle *et al.* 2018). These values for the baseline wind turbine intersect the BEMT curve and thus validate the code's accuracy in predicting the performance of the Greater Gabbard wind turbine. MSES, the MATLAB BEMT code, and the experimental conditions of Greater Gabbard have been validated and thus can be used for testing the multi-element design.

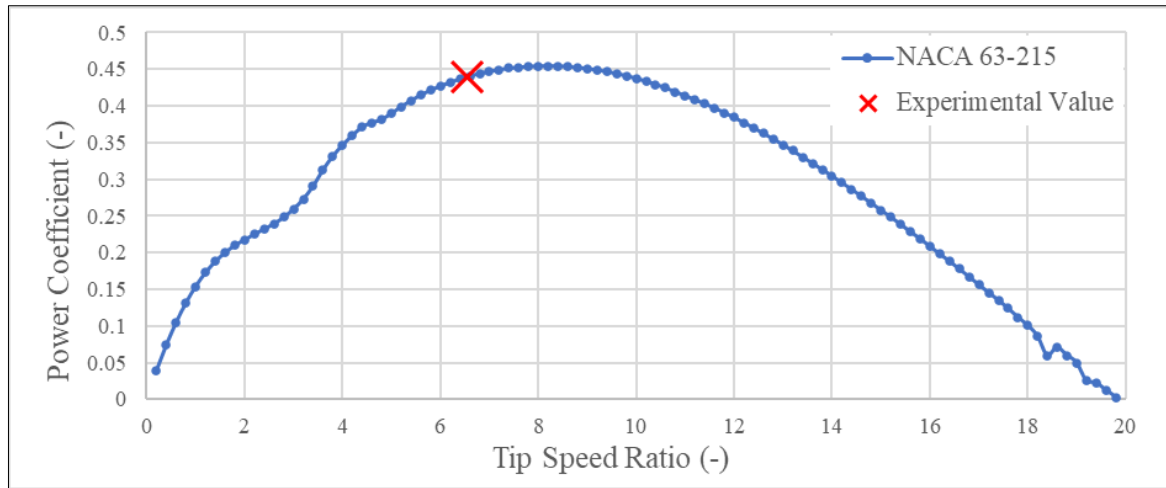


Figure 29: Coefficient of power against tip speed ratio performance curve predicted through BEMT code, with intersecting highlighted maximum mean power coefficient and design tip speed ratio for Greater Gabbard HAWT ($Re = 5 \times 10^6$, $C_{P,max} = 0.44$, Design TSR = 6.56) (Argyle *et al.* 2018).

7. Results and Discussion

7.1 Predicted Power Coefficient

The NACA 63-215 multi-element airfoil design was tested at the different configurations outlined in Table 3 and at the environmental conditions outlined in Table 4. The second element was deflected in 1° increments to determine the configuration for maximum power coefficient at different tip speed ratios as described in Section 6.2. The full set of data taken from MSES is shown in Appendix E-4. This data was then inputted into the MATLAB BEMT code to output the predicted performance curve at each configuration. The results from each configuration were gathered and are shown in Figure 30. For the varying second element, the maximum power coefficient achieved at different tip speed ratios was recorded and compared to a predicted performance curve of the single-element SWT-3.6-120 as shown in Figure 31.

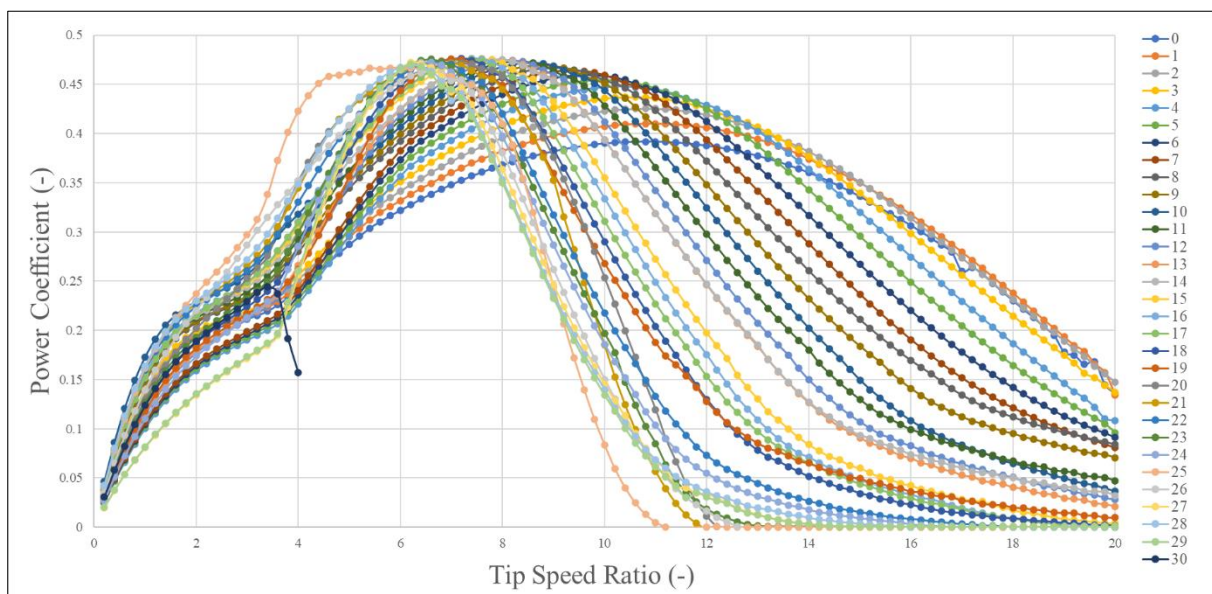


Figure 30: Accumulated power coefficient versus tip speed ratio curves for the incremental changes in angle of the second element of the multi-element HAWT blade ($Re = 5 \times 10^6$, Design TSR = 6.56).

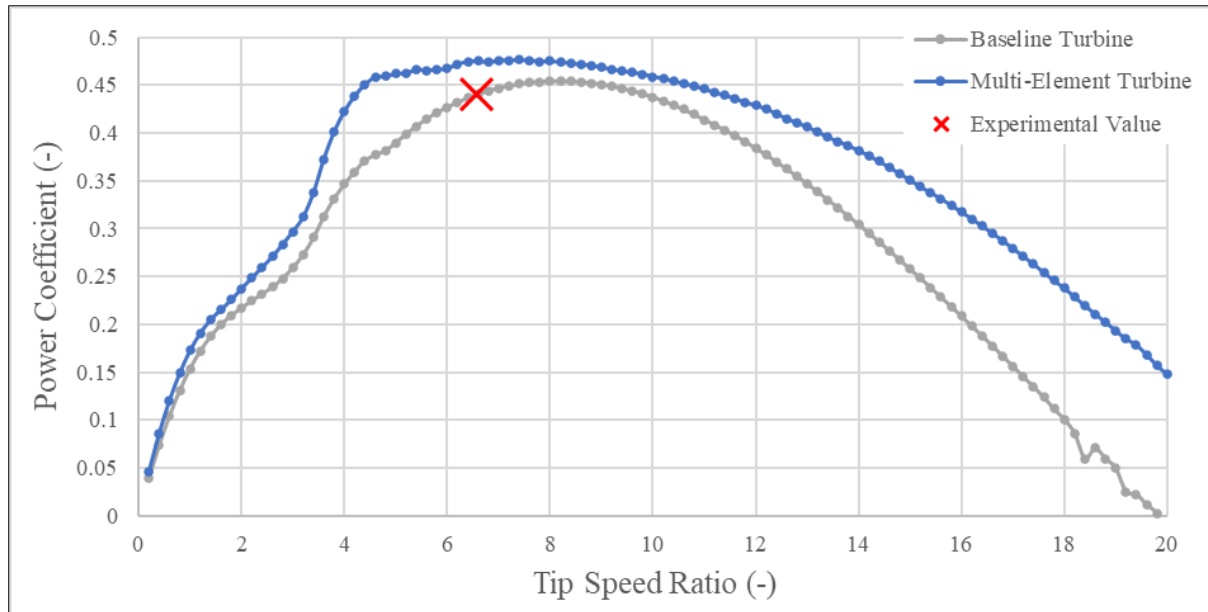


Figure 31: Power coefficient versus tip speed ratio curve showing the highest coefficient of power achieved by varying the second element's angle with wind speed and showing the predicted performance curve of the baseline SWT-3.6-120 HAWT at Greater Gabbard ($Re = 5 \times 10^6$, Design TSR = 6.56).

Evidently, the multi-element wind turbine blade achieves a greater power coefficient than that predicted for the baseline turbine. The multi-element wind turbine blade achieved a maximum power coefficient of 0.4763, which is approximately an 8.24% increase in maximum power coefficient compared to that of the SWT-3.6-120 HAWT at Greater Gabbard. This occurs at a tip speed ratio of 7.6. The percentage increase in power coefficient at different tip speed ratios is illustrated in Figure 32 and shows that the performance is significantly greater at higher tip speed ratios. This indicates that there will be an improvement in overall power output for the multi-element turbine design.

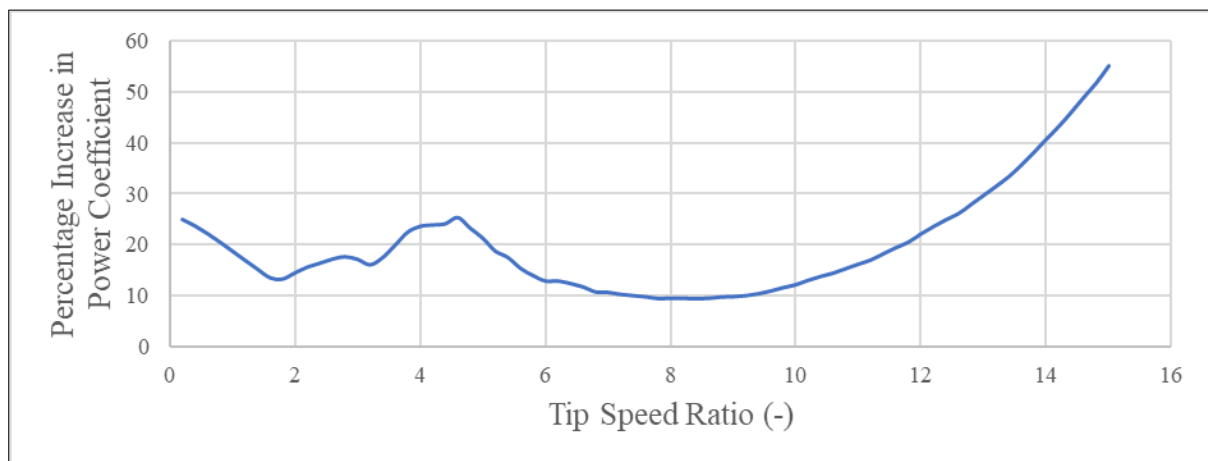


Figure 32: Percentage increase in power coefficient versus tip speed ratio curve showing the increase in power coefficient for the multi-element turbine compared to the baseline SWT-3.6-120 HAWT at Greater Gabbard ($Re = 5 \times 10^6$, Design TSR = 6.56).

7.2 Element Deflection

The second element must adjust with wind speed to achieve this power coefficient curve. The angle of the second element in each multi-element configuration required to achieve this performance was taken at different wind speeds and is shown below in Figure 33.

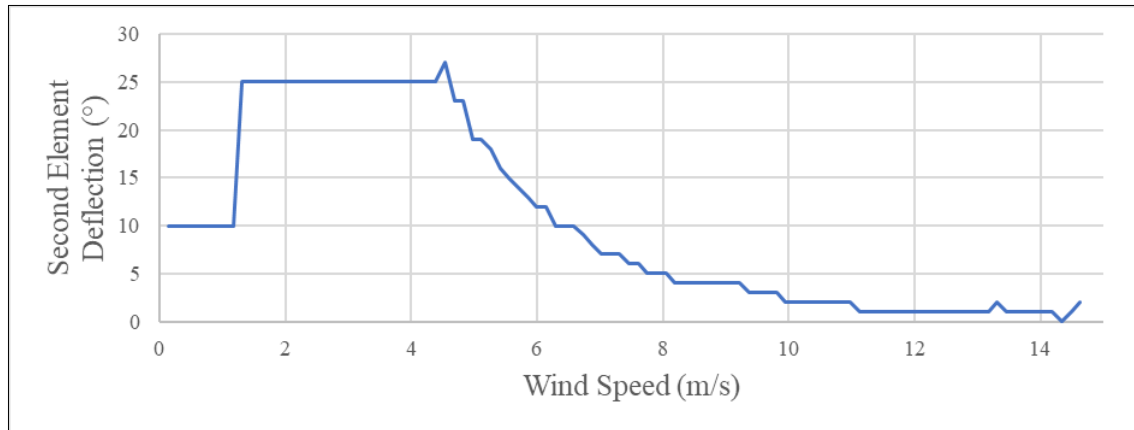


Figure 33: Second element deflection (°) versus wind speed (m/s) for the multi-element HAWT configuration.

At low wind speeds past the cut-in wind speed (i.e. wind speeds between 3 and 6 m/s), the second element is at a 25° angle. Beyond this the deflection steadily declines with increasing wind speed to decrease the lift and drag characteristics that were required to start the rotation of the turbine at low wind speeds. Figure 34 shows the distribution of second element deflection in 5° increments. This highlights the angles of deflection used most when the multi-element turbine blade is operating over the range of wind speeds. From this, the incremental change in deflection should be manufactured to be most accurate in the range of 0°-10° to achieve the desired performance.

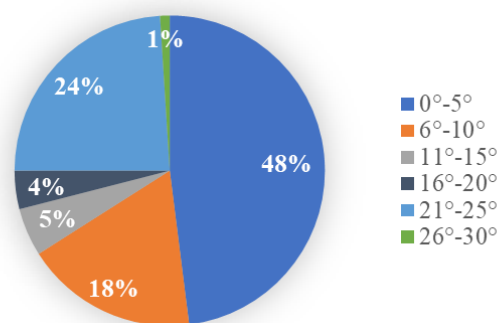


Figure 34: Percentage distribution of second element deflection angles for multi-element turbine.

7.3 Predicted Power Output

The expected power output at different wind speeds can be calculated by taking the maximum power coefficient for the multi-element airfoil configuration, with varying second element deflection, at different wind speeds and inputting the values into Equation 37 at the same environmental conditions as the baseline wind turbine. Figure 35 shows the power output of the multi-element wind turbine against the experimental data taken from the baseline wind turbine at Greater Gabbard, with a maximum power output of approximately 3.765 MW in comparison to the baseline 3.6 MW at rated wind speed (Argyle *et al.* 2018) (The Wind Power 2018). This is an improvement in power output of approximately 4.583%. The multi-element

design is predicted to outperform the baseline turbine at most wind speeds. Wind speeds where the baseline turbine outperforms the multi-element design are limited and may be due to the array of turbines tested at Greater Gabbard receiving varied wind speeds and the blockages caused by other turbines at the wind farm.

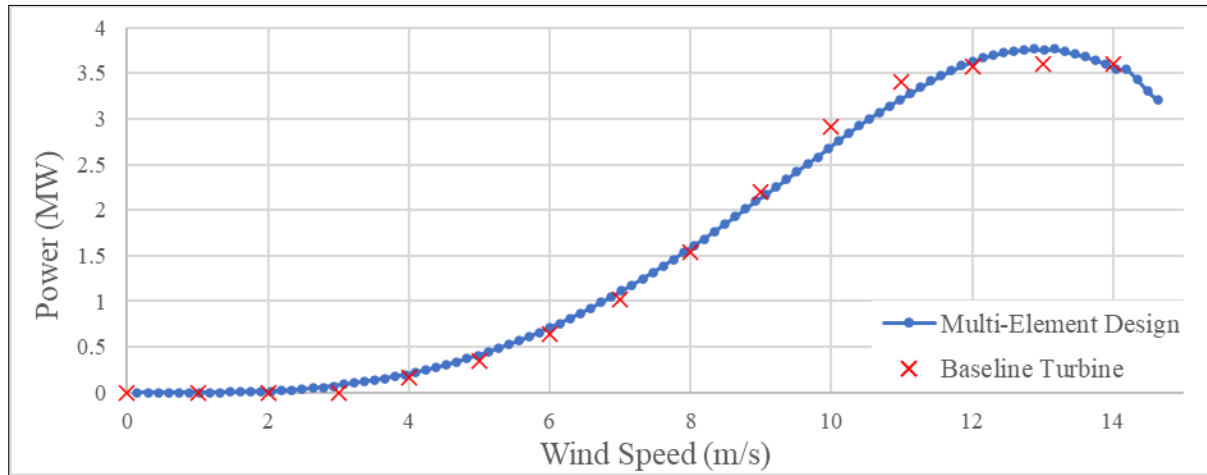


Figure 35: Power (MW) versus wind speed (m/s) showing the performance of the baseline project (SWT-3.6-120 at Greater Gabbard) and the multi-element wind turbine (Argyle *et al.* 2018) (The Wind Power 2018).

From Figure 35, it is seen that after the rated wind speed that the experimental data levels at 3.6 MW whereas the predicted performance of the multi-element turbine experiences stall-delay (Danmei *et al.* 2006). BEMT underpredicts the power output for a wind turbine in stalled condition when compared to measured performances at low tip speed ratios. A pitch-controlled wind turbine will adjust the turbine blades to stabilise aerodynamic efficiency at the rated wind speed and avoid stall delay (Mihet-Popa and Groza 2010). The SWT-3.6-120 turbine has pitch regulation with variable speed (Siemens 2011). This is not accounted for in the BEMT code and explains the change in behaviour between experimental and predicted power output after the rated wind speed. The predicted behaviour assumes that the turbine acts as a constant speed stall wind turbine that maintains a constant rotational speed regardless of wind speed (Polinder *et al.* 2003). The difference in power output between the two types of wind turbines is illustrated in Figure 36. Appendix E-5 shows an estimate for the predicted power output curve if the power output stabilised at the rated wind speed by adjusting pitch.

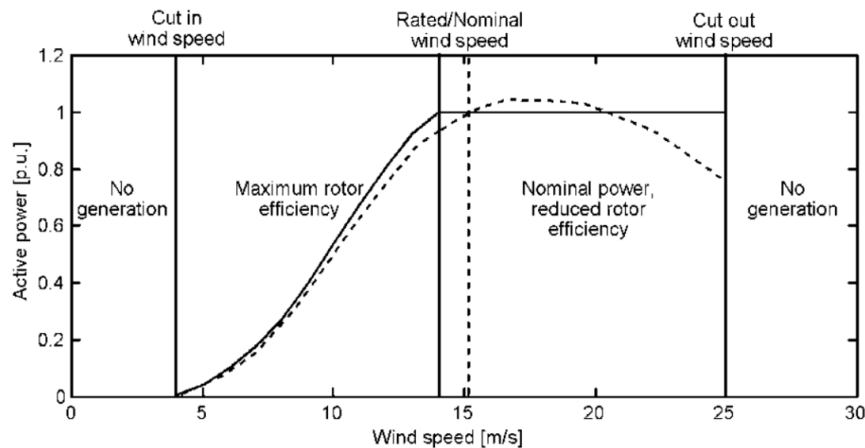


Figure 36: "Typical power curve of a constant speed stall (dotted) and a variable speed pitch (solid) controlled wind turbine" (Polinder *et al.* 2003).

7.4 Predicted Energy Yield

The energy yield for the multi-element HAWT can be predicted by inputting the predicted power output at different wind speeds into Equation 38, for the known wind speed frequency distribution at Greater Gabbard (see Figure 7) for a certain period of time. The results in Table 5 show the predicted energy yield for a four-week period comparing the baseline turbine at Greater Gabbard and the pitch-controlled multi-element turbine.

Table 5: Energy yield and power output comparison of baseline turbine to multi-element turbine for 4-week period.

Turbine	Maximum Power Output	Increase in Power Output	Energy Yield (4-weeks)	Increase in Energy Yield
SWT-3.6-120 (Baseline)	3.6 MW	-	251.28 GWh	-
Multi-element HAWT	3.765 MW	4.58%	272.9 GWh	8.60%

7.5 Chord and Twist Distribution

In addition to outputting the graph for power coefficient against tip speed ratio, the MATLAB BEMT code outputs the optimal chord and twist distribution for the inputted turbine blade. Figure 37 and Figure 38 show the predicted optimal chord and twist distribution along the span of the multi-element wind turbine blade. The baseline wind turbine at Greater Gabbard has a root chord of 4.2 metres and a tip chord of 1 metre (Siemens 2012). These values are marked in Figure 37 to highlight the feasibility of physically creating these turbine blades. The root, peak, and tip chord and twist values for the multi-element blade design are shown in Table 6. Although these distributions are not under investigation in this project, they have typical values for blades approximately 60 metres long and therefore the multi-element blade has a feasible design. The chord distribution is similar to the baseline blade and the twist distribution is similar to turbines of similar airfoil studies as highlighted in Appendix E-6 (Taher *et al.* 2018).

Table 6: Chord and twist for multi-element blade design.

Position	Chord (m)	Twist (°)
Root	4.277	17.696
Peak	6.141	29.240
Tip	1.071	-0.227

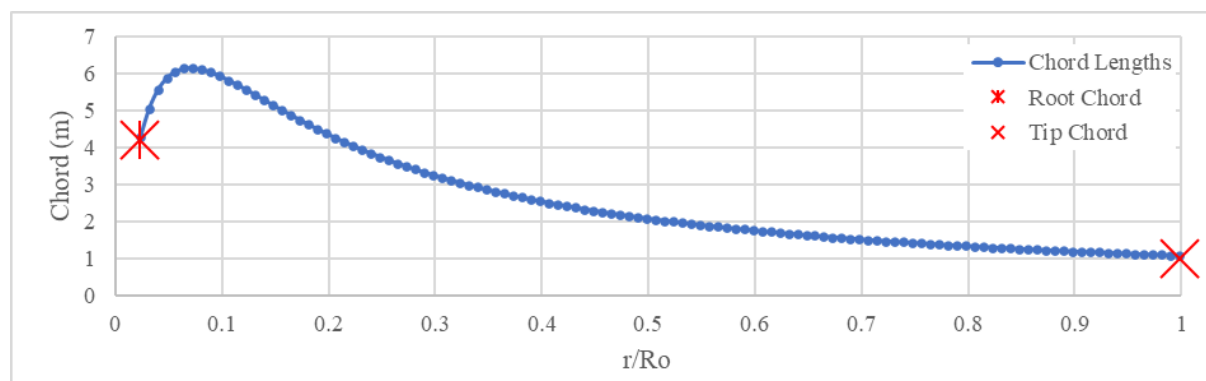


Figure 37: Chord distribution for multi-element configuration of NACA 63-215 with baseline root and tip chord highlighted to show the feasibility of manufacturing the multi-element wind turbine blade (Siemens 2012).

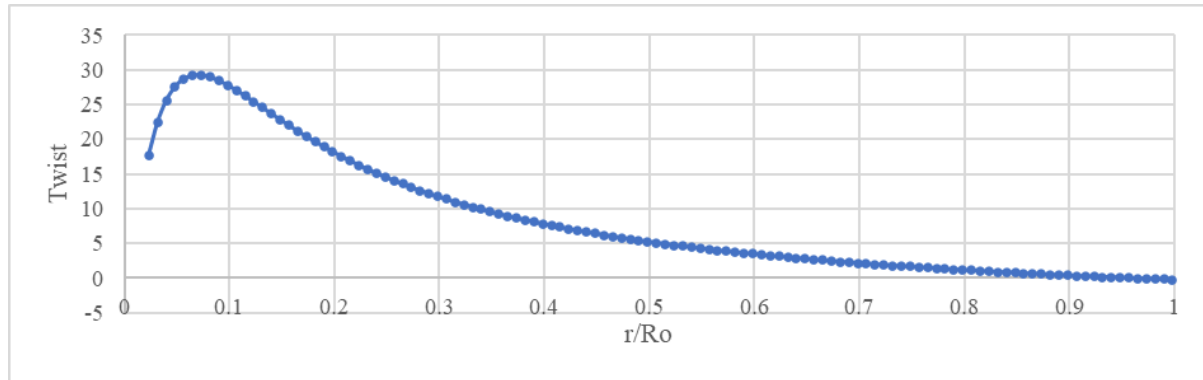


Figure 38: Twist distribution along the blade of the multi-element configuration of NACA 63-215.

8. Conclusions and Recommendations

This project discussed the effect of incorporating a multi-element airfoil HAWT blade into a pre-existing wind turbine design. The SWT-3.6-120, located at Greater Gabbard wind farm, was chosen for the availability of experimental power output data and its likeness to wind turbines and environmental conditions in Ireland. The multi-element design led to a 4.58% increase in maximum power output and an 8.6% increase in energy yield. Due to the cost associated with multi-element blades, this relatively minimal increase in performance may indicate that the technology should be improved before multi-element blades are tested on full-scale HAWTs. Wind energy is an ever-growing sector of renewable energy technologies that will greatly reduce the negative effect humans have had on the planet. Improvements in this field of study are vital in order to reach the goals set out by the Climate Action Plan for 2050.

Investigation into the following topics would give further insight into the capabilities of multi-element blade design:

1. Investigate multi-element configurations that incorporate more than two elements to determine the effect on turbine efficiency.
2. Investigate the use of different airfoils within the configuration under analysis, in a similar way to the NLR-7301 airfoil tested at NLR.
3. Run a CFD analysis of the multi-element airfoil configuration used in this project to further verify the aerodynamic characteristics investigated in this project and determine other three-dimensional aerodynamic characteristics.
4. Investigate a blade design similar to the blade shown in Figure 5, where the blade incorporates the multi-element design along the mid-span of the blade but has a single-element airfoil configuration at the root and tip.
5. Improvement of the MATLAB BEMT code used within this project to allow the input of multiple airfoil profiles along the span of the blade as seen in existing turbines.

References

Ahmed, T., Amin, M.T., Islam, S.R. and Ahmed, S., (2014) ‘Computational study of flow around a NACA 0012 wing flapped at different flap angles with varying Mach numbers’, *Glob J Res Eng*, 13(4), pp.4-16.

Airfoil Tools (2024) *Airfoil Tools*, available: <http://airfoiltools.com/> [Accessed 2 Feb. 2024].

Argyle, P., Watson, S., Montavon, C., Jones, I. and Smith, M. (2018) ‘Modelling turbulence intensity within a large offshore wind farm’, *Wind Energy*, 21(12), pp.1329-1343.

Argyle, P., Watson, S., Montavon, C., Jones, I. and Smith, M. (2015) ‘Turbulence intensity within large offshore wind farms’.

Bah, A., Sankar, L.N., and Jagoda, J.I. (2013) ‘Numerical Investigations on the Use of Multi-Element Airfoils for Vertical Axis Wind Turbine Configurations’, ASME 2013 Gas Turbine India Conference.

Bertagnolio, F., Sørensen, N., Johansen, J. and Fuglsang, P. (2001) ‘Wind turbine airfoil catalogue.

Blakeley, A., Flay, R., and Richards, P. (2012) ‘Design and Optimisation of Multi-Element Wing Sails for Multihull Yachts’, 18th Australasian Fluid Mechanics Conference Launceston.

Burton, T., Jenkins, N., Sharpe, D. and Bossanyi, E. (2011) ‘Wind energy handbook’, John Wiley & Sons.

Chaviaropoulos, P.K. and Hansen, M.O. (2000) ‘Investigating three-dimensional and rotational effects on wind turbine blades by means of a quasi-3D Navier-Stokes solver’, *J. Fluids Eng.*, 122(2), pp.330-336.

Da Rosa, A.Vieira. (2009) ‘Fundamentals of Renewable Energy Processes’, Second edition, Amsterdam ; Elsevier Academic Press.

Danmei Hu, Ouyang Hua, Zhaohui Du (2006) ‘A study on stall-delay for horizontal axis wind turbine’, *Renewable Energy*, Volume 31, Issue 6, pp 821-836

Das, A., Chimonyo, K.B., Kumar, T.R., Gourishankar, S. and Rani, C. (2017) ‘Vertical axis and horizontal axis wind turbine-A comprehensive review’, In 2017 international conference on energy, communication, data analytics and soft computing, pp. 2660-2669

Drela, M. (2007) ‘A User’s Guide to MSES 3.05’, Massachusetts Institute of Technology (MIT), Cambridge.

EI. (2023), “The Energy Institute (EI) Statistical Review of World Energy”, Energy Institute, 72, p3 p49

Energia (2017) ‘Wind Energy Ireland: Top 10 Facts’ Energia Group, available: <https://energiagroup.com/news-media/wind-energy-ireland-top-10-facts/> [accessed 28 Nov 2023].

Energia (2017) ‘Wind Energy Ireland: Top 10 Facts’, Energia Group, available: <https://energiagroup.com/news-media/wind-energy-ireland-top-10-facts/> [accessed 28 Nov 2023].

Energy.gov. (2023) *Airfoils, Where the Turbine Meets the Wind*, available: <https://www.energy.gov/eere/wind/articles/airfoils-where-turbine-meets-wind> [Accessed 23 Mar. 2024].

European Commission (2020) ‘Causes of Climate Change’, Climate Action, available: https://climate.ec.europa.eu/climate-change/causes-climate-change_en [accessed 28 Nov 2023].

Fernandes, A., Alves, J., Neto, O., Antunes, A., Teixeira De Mendonça, M., Luiz, J. and Azevedo, F. (2005) ‘Numerical study of two-dimensional high-lift configurations using the MSES code’, In Proceedings of the 18th International Congress of Mechanical Engineering, COBEM 2005.

GE Wind Energy (2003) ‘Arklow Bank Wind Park 25 MW Offshore Wind Power Facility’, available: <https://www.mresearch.com/pdfs/docket4185/NG11/doc32.pdf>. [accessed 28 Nov 2023].

Ge, M., Fang, L. and Tian, D. (2015) ‘Influence of Reynolds Number on Multi-Objective Aerodynamic Design of a Wind Turbine Blade’, PloS one, 10(11), p.e0141848.

Government of Ireland (2023) *Climate Action Plan 2023*, Government of Ireland, available: <https://www.gov.ie/pdf/?file=https://assets.gov.ie/270956/94a5673c-163c-476a-921f-7399cdf3c8f5.pdf#page=null>. [accessed 28 Nov 2023].

Hansen, M. (2015) ‘Aerodynamics of wind turbines’, Routledge.

Het Bhavsar, Sukanta Roy, Hakeem Niyas (2023) ‘Multi-element airfoil configuration for HAWT: A novel slot design for improved aerodynamic performance’, Materials Today: Proceedings, Volume 72, Part 1, Pages 386-393.

Jonkman, J. (2014) *NREL 5MW Rotor Speed Data*, NREL Forum, available: <https://forums.nrel.gov/t/nrel5mw-rotor-speed-data/867> [Accessed 15 Mar. 2024].

Jonkman, J., Butterfield, S., Musial, W. and Scott, G. (2009) ‘Definition of a 5-MW reference wind turbine for offshore system development (No. NREL/TP-500-38060)’, National Renewable Energy Lab.(NREL), Golden, CO (United States).

Kuang, L., Li, H., Su, X., Song, X., Wang, Z., Zhang, Y. and Ma, W. (2023) ‘Effect of chord length ratio on aerodynamic performance of two-element wing sail’, Ocean Engineering, 287, p.115946.

Ledoux, J., Rizzo, S. and Salomon, J. (2021) ‘Analysis of the Blade Element Momentum Theory’, SIAM Journal on Applied Mathematics, 81(6), pp.2596–2621.

Mahmuddina, F. (2016) ‘The effect of flat plate theory assumption in post-stall lift and drag coefficients extrapolation with Viterna method’, Science and Engineering, 6, pp.9-13.

Manwell, J. F., McGowan, J. G. and Rogers, A. L. (2010) 'Wind Energy Explained', 2nd ed., United Kingdom: John Wiley & Sons Ltd.

Maughmer, M.D. and Coder, J.G. (2010) 'Comparisons of theoretical methods for predicting airfoil aerodynamic characteristic', US Army Aviation Research, Development and Engineering Command RDECOM TR.

Met Éireann (2023) 'Wind over Ireland', available: <https://www.met.ie/climate/what-we-measure/wind#:~:text=Average%20annual%20wind%20speeds%20range,locations%20such%20as%20Malin%20Head> [accessed 28 Nov 2023].

Mihet-Popa, Lucian & Voicu, Groza (2010) 'Dynamic modeling, simulation and control strategies for 2 MW wind generating systems', International Review on Modelling and Simulations. 3. 1410-1418.

MIT (2024) *MSES - Software for High Lift Multielement Airfoil Configurations* | MIT Technology Licensing Office, available: <https://tlo.mit.edu/industry-entrepreneurs/available-technologies/mses-software-high-lift-multielement-airfoil> [Accessed 11 Feb. 2024].

Moriarty, P.J. and Hansen, A.C. (2005) 'AeroDyn theory manual', (No. NREL/TP-500-36881), National Renewable Energy Lab., Golden, CO (US).

National Geographic (2023) *Global Warming*, available: <https://education.nationalgeographic.org/resource/global-warming/> [accessed 28 Nov 2023].

NREL (2016) *NREL_5MW_126_RWT.csv*, GitHub, available: https://github.com/NREL/turbine-models/blob/master/Offshore/NREL_5MW_126_RWT.csv [Accessed 15 Mar. 2024].

Polinder, Henk & Dubois, Maxime & Slootweg, J. (2003) 'Generator systems for wind turbines', Proceedings PCIM'03, 6.

R. Lanzafame, M. Messina (2007) 'Fluid dynamics wind turbine design: Critical analysis, optimization and application of BEM theory', Renewable Energy, Volume 32, Issue 14, pp.2291-2305.

Raftery, A. (2023) 'A Critical Review of Wind Energy Usage in Ireland', Submitted to the University of Limerick.

Ragheb, A.M. and Selig, M.S. (2017) 'Multielement airfoils for wind turbine', In Wind Energy Engineering (pp. 203-219). Academic Press.

Ruhland, J. and Breitsamter, C. (2021) 'Numerical analysis of high-lift configurations with oscillating flaps', CEAS Aeronautical Journal, 12, pp.345-359.

SEAI (2023) 'Energy in Ireland 2022 Report', Sustainable Energy Authority of Ireland.

Series NF (2012) 'The NACA airfoil series'.

Siemens (2011) 'Thoroughly tested, utterly reliable Siemens Wind Turbine SWT-3.6-120'.

Siemens (2012) ‘Siemens Power Generation’, available: <https://assets.gov.ie/110510/5f331d82-a5fe-47e3-8af5-bf913f5cd5bc.pdf> [Accessed 14 Feb. 2024].

Silva, A., Neto, J., Antunes, A., Mendonça, M., Azevedo, J., Neto, A. (2005) ‘Numerical Study of Two-Dimensional High-Lift Configurations using the MSES Code’, in Proceedings of COBEM 2005, Presented at the 18th International Congress of Mechanical Engineering,

SSE Renewables (2016) *Greater Gabbard* | *SSE Renewables* available: <https://www.sserenewables.com/offshore-wind/operational-wind-farms/greater-gabbard/> [Accessed 20 Feb. 2024].

SSE Renewables (2017) *Galway Wind Park*, SSE Renewables, available: <https://www.sserenewables.com/onshore-wind/ireland/galway-wind-park/#:~:text=The%20174MW%20Galway%20Wind%20Park,generation%20site%20on%20the%20island> [accessed 28 Nov 2023].

SSE Renewables (2023) *Arklow Bank Wind Park*, SSE Renewables, available: <https://www.sserenewables.com/offshore-wind/projects/arklow-bank-wind-park/> [accessed 28 Nov 2023].

Taher, Ahmed & Elgabaili, Mohamed & Rajab, Zakariya & Buaossa, Nagi & Khalil, Ashraf & Mohamed, Faisal (2018) ‘Optimization of small wind turbine blades using improved blade element momentum theory’, *Wind Engineering*, 43(3), pp.299-310.

Tahir, Ahmed & Elgabaili, Mohamed & Rajab, Zakariya & Buaossa, Nagi & Khalil, Ashraf & Mohamed, Faisal (2018) ‘Optimization of small wind turbine blades using improved blade element momentum theory’, *Wind Engineering*, 43(3), pp.299-310.

The IRC (2023) ‘10 Countries at Risk of Climate Disaster’, available: <https://www.rescue.org/article/10-countries-risk-climate-disaster> [accessed 28 Nov 2023].

The Wind Power (2018) ‘Siemens SWT-3.6-120 - Manufacturers and turbines’, available: https://www.thewindpower.net/turbine_en_79_siemens_swt-3.6-120.php [Accessed 14 Feb. 2024].

Westwood, Rachel (2012), ‘Seismic monitoring and multiphysics modelling of ground-borne vibrations from small wind turbines’.

Wind Energy Ireland (2023) *Ireland Now Number One in the World for Onshore Wind Energy*, Wind Energy Ireland.

Appendices

Appendix A – Study: Multi-Element Airfoil Scale Factor

Testing was performed to determine the best-performing scale factor for the NACA 63-215 multi-element airfoil configuration, as discussed in Section 6.2. A study performed by Kuang *et al.* (2023) tested a multi-element airfoil configuration for a number of different scale factors for a NACA 0020 airfoil. Testing was performed at the three best performing scale factors for each element to determine the scale factor to be used in this project for the NACA 63-215. Figure 39 shows the three configurations considered.

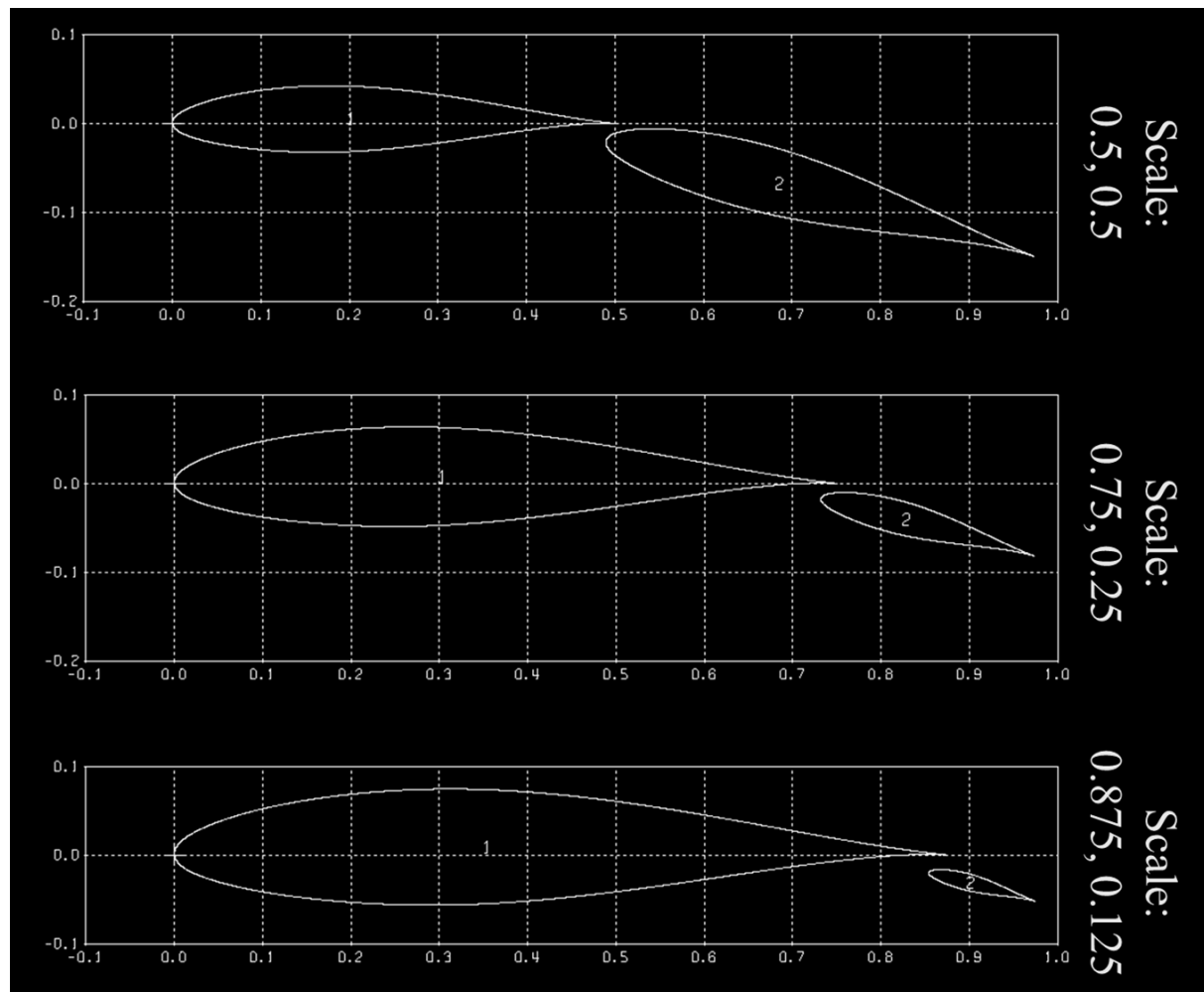


Figure 39: NACA 63-215 multi-element configurations with scale factors of (0.5, 0.5), (0.75, 0.25), (0.875, 0.125) respectively.

Each of these configurations were tested in MSES and inputted into the MATLAB BEMT code to determine their power coefficient versus tip speed ratio curve. The configurations were tested with the second element deflecting from 0° - 25° in 5° increments. The results from these are shown in the figures below. The maximum power coefficient achieved at each wind speed for the varying second element was recorded for the configurations. Figure 43 shows the results of this. By calculating the area underneath each curve for the operating wind speeds it was determined that the “0.75, 0.25” scale factor configuration for the first and second element respectively would provide the best performing multi-element wind turbine blade for the

NACA 63-215 as it had the maximum power coefficient across the most wind speeds. The summations for the power coefficients under the curve for each scale factor configuration are 4.113786, 4.382573, and 4.313814 respectively. Once the optimum scale factor was chosen, the spacing was calculated for the multi-element configuration based off of other works to have the optimum lift/drag ratio that could function in the software used (Silva *et al.* 2005).

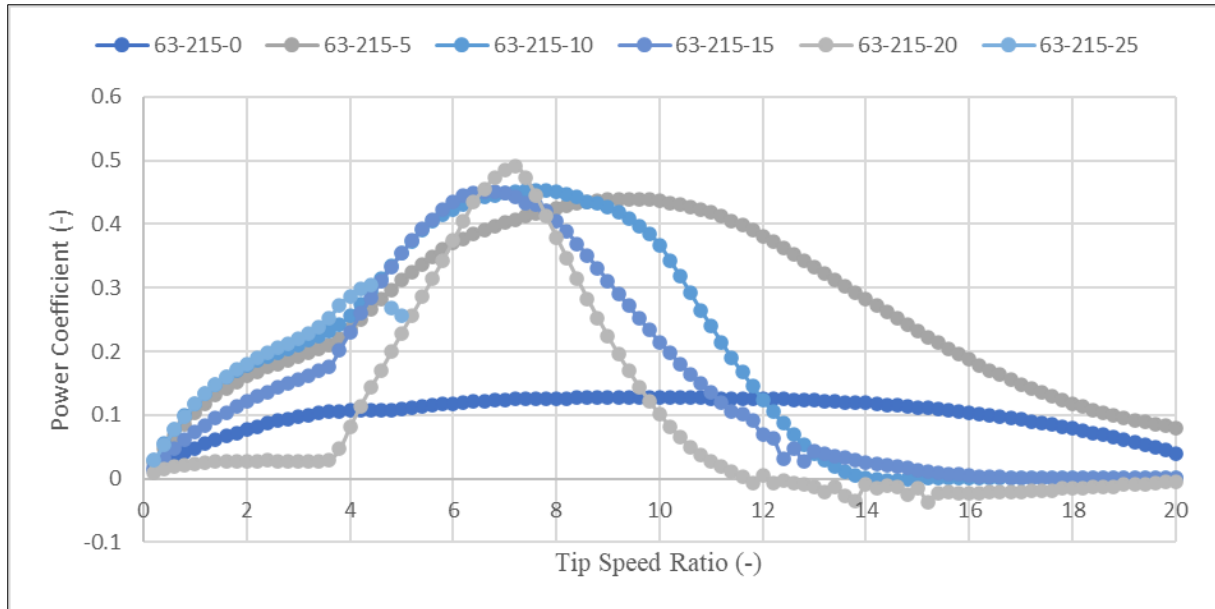


Figure 40: Accumulated power coefficient versus tip speed ratio curves for the incremental changes in angle for the second element of the multi-element HAWT blade with a 0.5, 0.5 scale factor for the first and second element respectively ($Re = 5 \times 10^6$, Design TSR = 6.56).

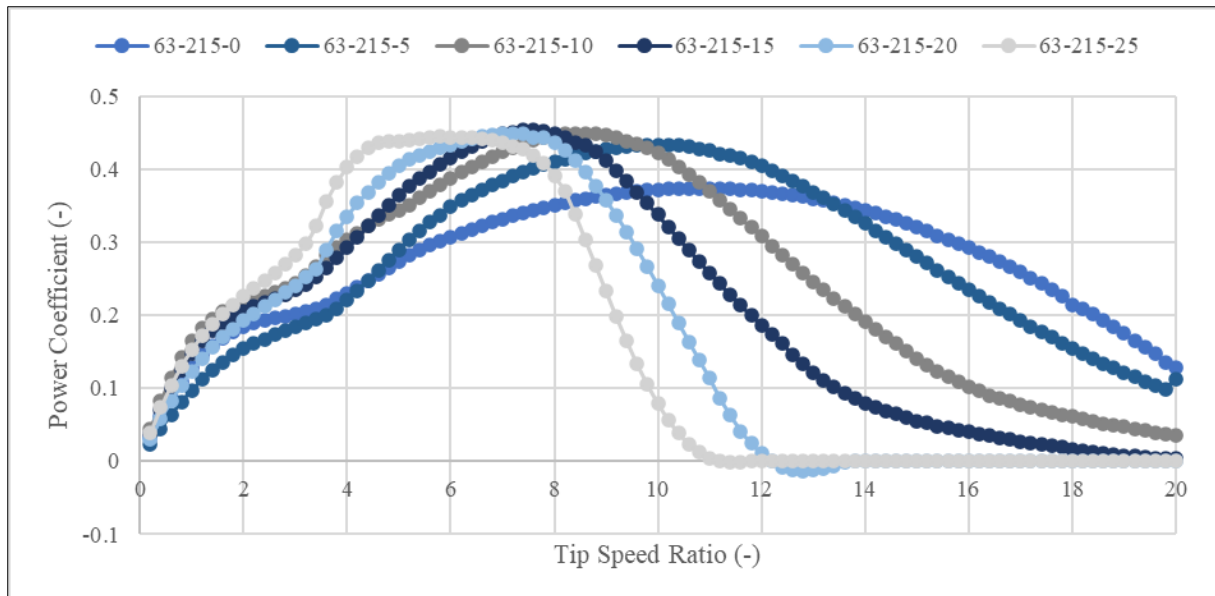


Figure 41: Accumulated power coefficient versus tip speed ratio curves for the incremental changes in angle for the second element of the multi-element HAWT blade with a 0.75, 0.25 scale factor for the first and second element respectively ($Re = 5 \times 10^6$, Design TSR = 6.56).

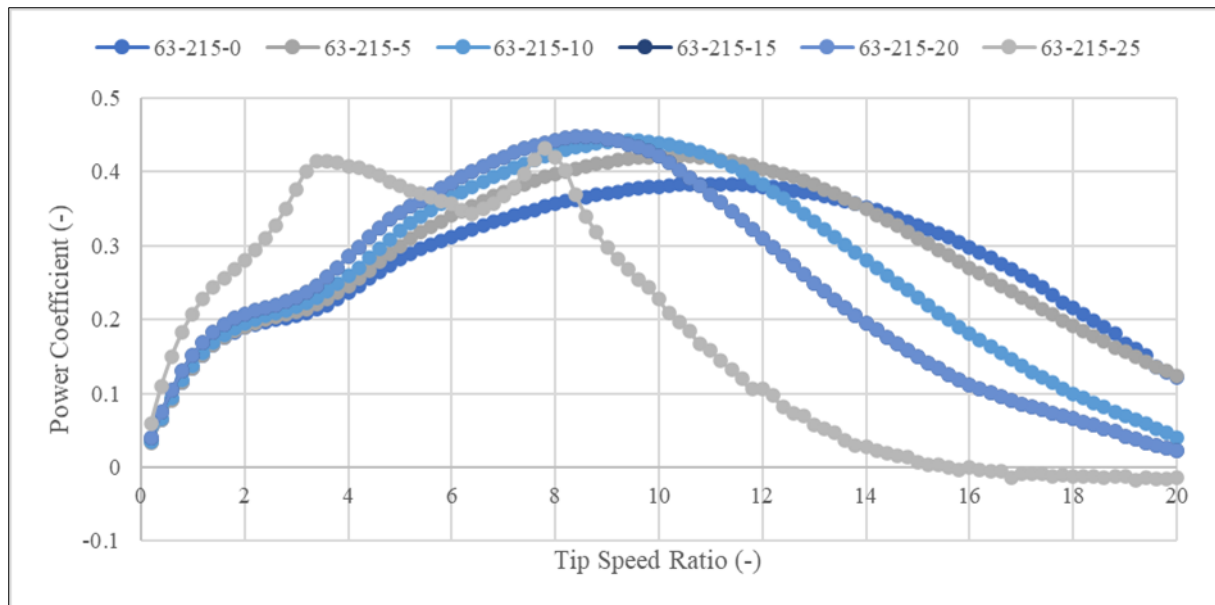


Figure 42: Accumulated power coefficient versus tip speed ratio curves for the incremental changes in angle for the second element of the multi-element HAWT blade with a 0.875, 0.125 scale factor for the first and second element respectively ($Re = 5 \times 10^6$, Design TSR = 6.56).

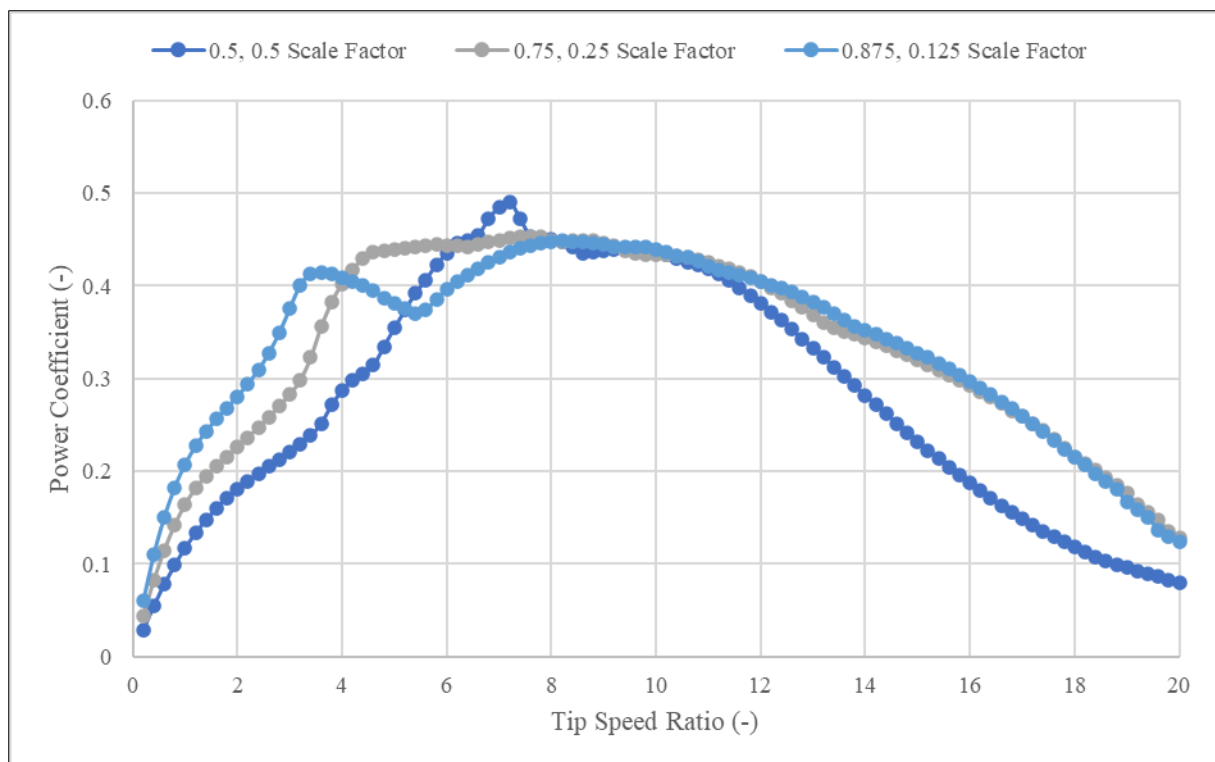


Figure 43: Accumulated power coefficient versus tip speed ratio curves for the three scale factor configurations for the elements of the multi-element HAWT blade ($Re = 5 \times 10^6$, Design TSR = 6.56).

Appendix B – Further Plots for Validation of MSES

For the study discussed in Section 5.1.1, Figure 23 shows the MSES study carried out in this project with experimental data points overlayed to highlight the accuracy of MSES for use in this project. The individual plots used to create Figure 23 are shown in Figure 44 and Figure 45 below.

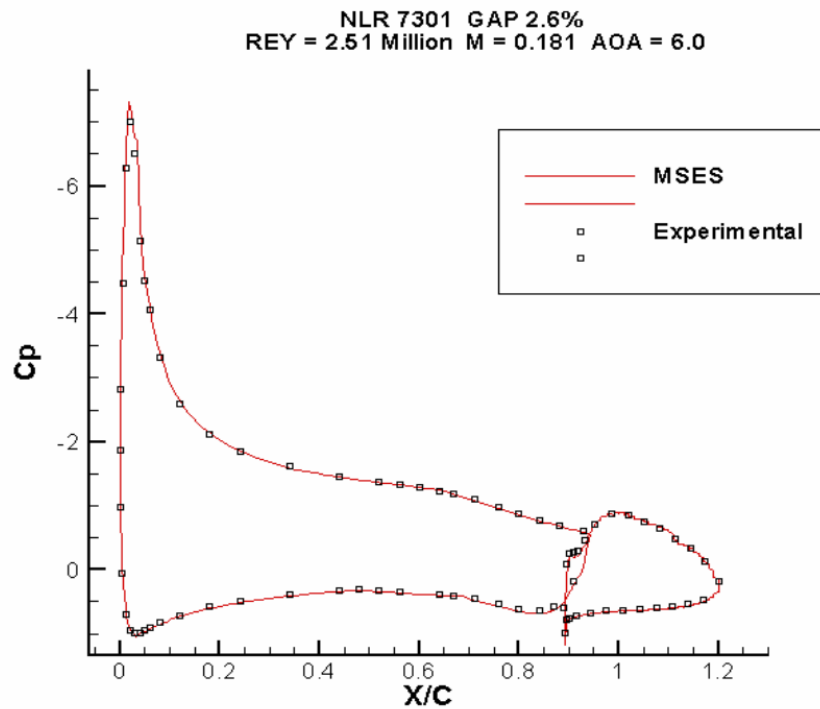


Figure 44: NLR study. MSES and experimental pressure distribution for multi-element NLR-7301 with $Re=2.51 \times 10^6$, $M=0.181$, $\alpha=6^\circ$ (Fernandes *et al.* 2005).

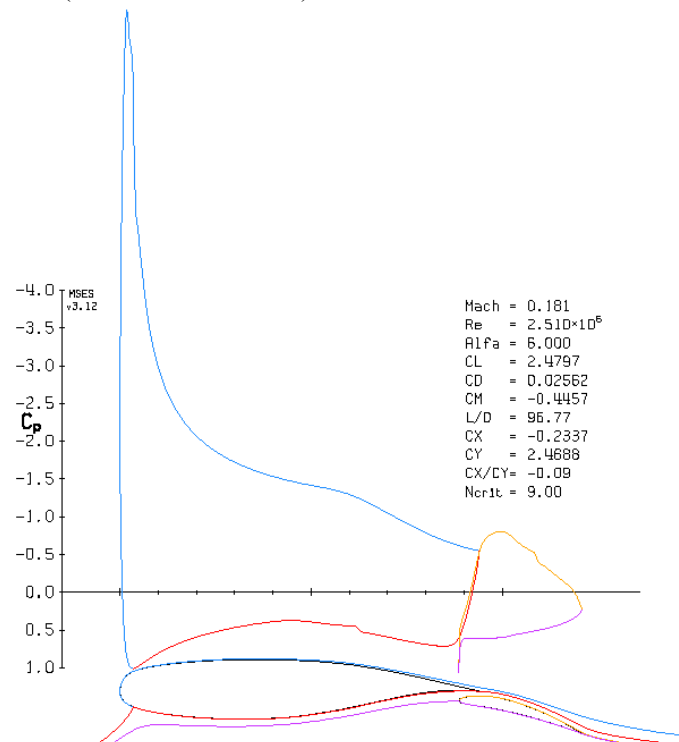


Figure 45: Local simulation study of MSES using parameters outlined by NLR, with $Re=2.51 \times 10^6$, $M=0.181$, $\alpha=6^\circ$ (Fernandes *et al.* 2005).

Appendix C – NREL 5 MW Experimental Data

The coefficient of power versus wind speed experimental data for the NREL 5 MW wind turbine discussed in Section 5.2.1 is shown in Figure 46. The full dataset used from Jonkman (2014) is shown in Table 7.

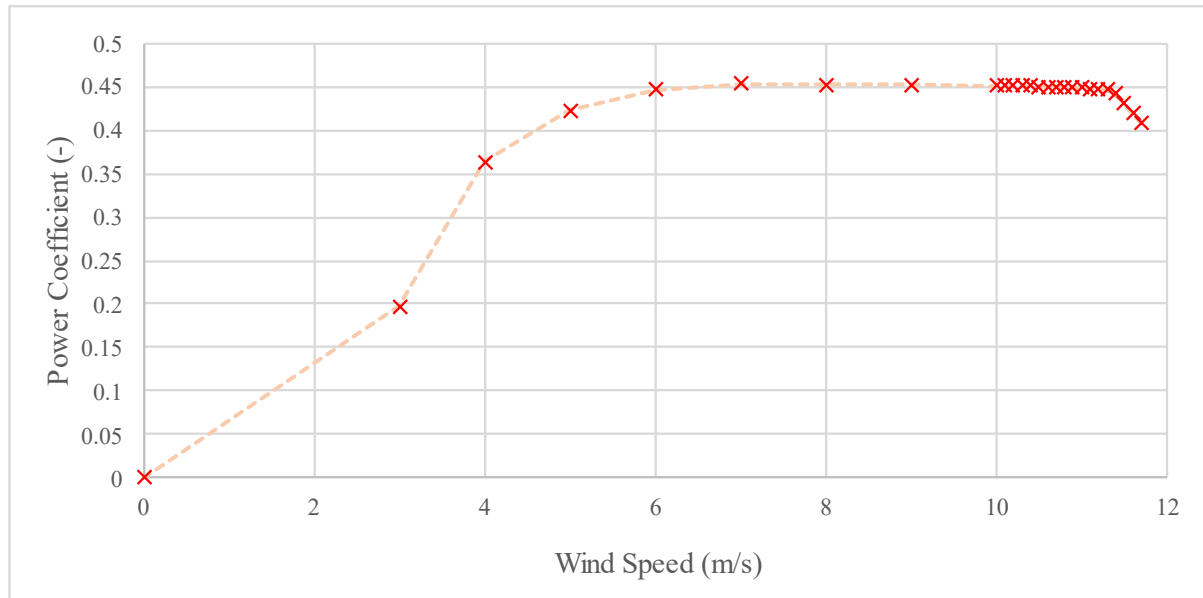


Figure 46: Coefficient of power against tip speed ratio performance curve of experimental data from NREL 5 MW wind turbine (Jonkman 2014) (NREL 2016).

Table 7: NREL 5 MW Experimental Data (Jonkman 2014) (NREL 2016).

Wind Speed [m/s]	Power [kW]	Cp [-]	Calculated TSR [-]
0	0	0	0
3	40.5	0.196405748	3.75
4	177.7	0.363555224	5
5	403.9	0.423084169	6.25
6	737.6	0.447126173	7.5
7	1187.2	0.453202924	8.75
8	1771.1	0.452935183	10
9	2518.6	0.452370844	11.25
10	3448.41	0.451525031	12.5
10.1	3552.15	0.451429682	12.625
10.2	3657.95	0.451336218	12.75
10.3	3765.16	0.451164269	12.875
10.4	3873.95	0.450938098	13
10.5	3984.49	0.450679475	13.125
10.6	4096.56	0.450365029	13.25
10.7	4210.69	0.450054254	13.375
10.8	4326.15	0.449669304	13.5
10.9	4443.41	0.449262148	13.625
11	4562.51	0.44883706	13.75

11.1	4683.43	0.448392183	13.875
11.2	4806.18	0.44792871	14
11.3	4929.92	0.447370638	14.125
11.4	5000.37	0.441926986	14.25
11.5	5000.02	0.430468281	14.375
11.6	5000	0.419429499	14.5
11.7	4999.99	0.408765737	14.625

A number of airfoils were modified and chosen by NREL to use on this turbine. Airfoil input data for the NACA 64_A17 was calculated by the NREL for the values for the lift and drag coefficients of the turbine at different angles of attack (Jonkman *et al.* 2009). This data was used as an input for an additional validation of the MATLAB BEMT code in this project as, the input data of which is shown below. C_L and C_D coefficients were taken from the study of the 5 MW turbine by Jonkman *et al.* (2009) and used to compare with the C_P performance curve outlined in Jonkman (2014) and NREL (2016) as shown in Figure 46.

α	C_L	C_D
0	0.442	0.0052
5	1.011	0.0058
6	1.103	0.0091
7	1.181	0.0113
8	1.257	0.0124
9	1.326	0.0136
10	1.382	0.0150
11	1.415	0.0383
12	1.434	0.0613
13	1.451	0.0841
14	1.448	0.1065
15	1.445	0.1287
20	1.428	0.2379

Appendix D – Further Experimental Conditions for Greater Gabbard

Figures used in determining parameters outlined in Section 6.2.1 are shown below.

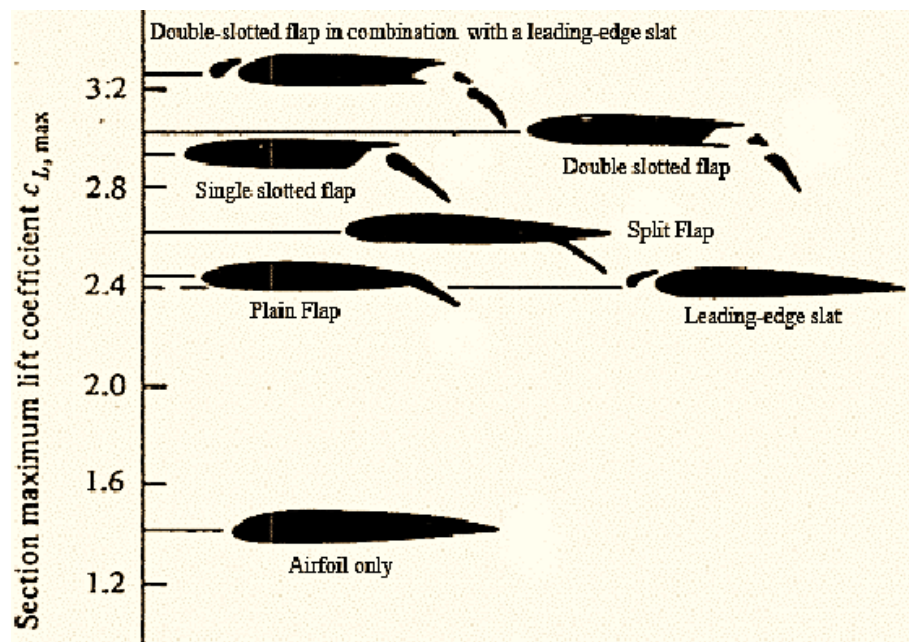


Figure 47: Lift coefficient values of various multi-element airfoils (Ahmed *et al.* 2013).

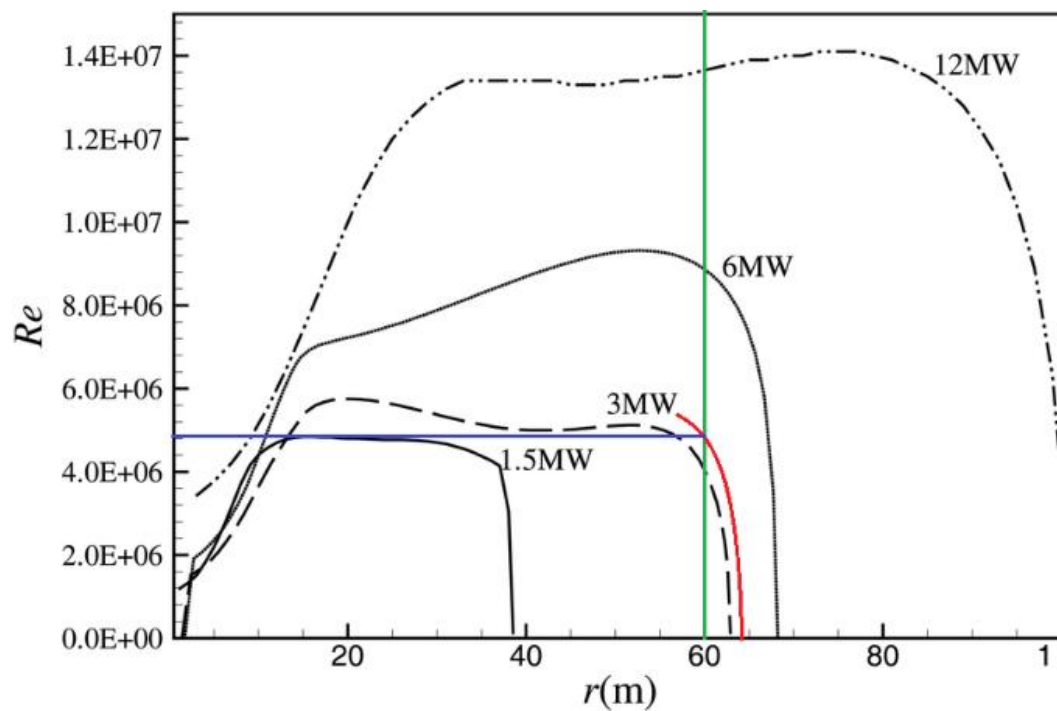


Figure 48: Reynolds number versus blade length for various MW HAWTs (Ge *et al.* 2015).

Appendix E – Additional Data from MSES and BEMT Code

Appendix E-1 – NACA 63-215 .Airfoil Profile

The text (.dat) file used as an input in MSES for the NACA 63-215 airfoil profile is shown below.

NACA 63-215 AIRFOIL

1	0
0.95014	0.00616
0.9003	0.01368
0.85043	0.02213
0.80051	0.03105
0.75055	0.04014
0.70053	0.04906
0.65047	0.05751
0.60035	0.06524
0.55019	0.07203
0.5	0.07768
0.44977	0.08194
0.39952	0.08457
0.34926	0.0853
0.299	0.08392
0.24875	0.08049
0.19852	0.07487
0.14834	0.06682
0.09823	0.05569
0.07323	0.04847
0.04829	0.0396
0.02348	0.02792
0.0112	0.0198
0.00637	0.01528
0.00399	0.0125
0	0
0.00601	-0.0115
0.00863	-0.01388
0.0138	-0.01766
0.02652	-0.0242
0.05171	-0.03328
0.07677	-0.03999
0.10177	-0.04535
0.15166	-0.05336
0.20148	-0.05895
0.25125	-0.06259
0.301	-0.06448
0.35074	-0.0647
0.40048	-0.06315
0.45023	-0.06004
0.5	-0.05562
0.54981	-0.05013
0.59965	-0.04382
0.64953	-0.03691
0.69947	-0.02962
0.74945	-0.02224
0.79949	-0.01513
0.84957	-0.00867
0.8997	-0.00334
0.94986	0.00016
1	0

Appendix E-2 – Sample MSES Input File

Below is a sample input file for MSES. Note that ALFAin changes with the angle of attack under consideration:

```

3 4 5 7
3 4 5 7
0.04000 0.20000 0.00000 | MACHin CLIFin ALFAin
3 2 0.1008 -0.0549 0.0000 | ISMOM IFFBC [ DOUXin DOUYin SRCEin ]
5.000E+06 2.000 1 | REYNin ACRIT [ KTRTYP ]
1.0000 1.0000 1.0000 1.0000 | XTR1 XTR2
0.970 -1.200 | MCRIT MUCON
0 0 | ISMOVE ISPRES
0 0 | NMODN NPOSN

```

Appendix E-3 – Sample MSES Plots

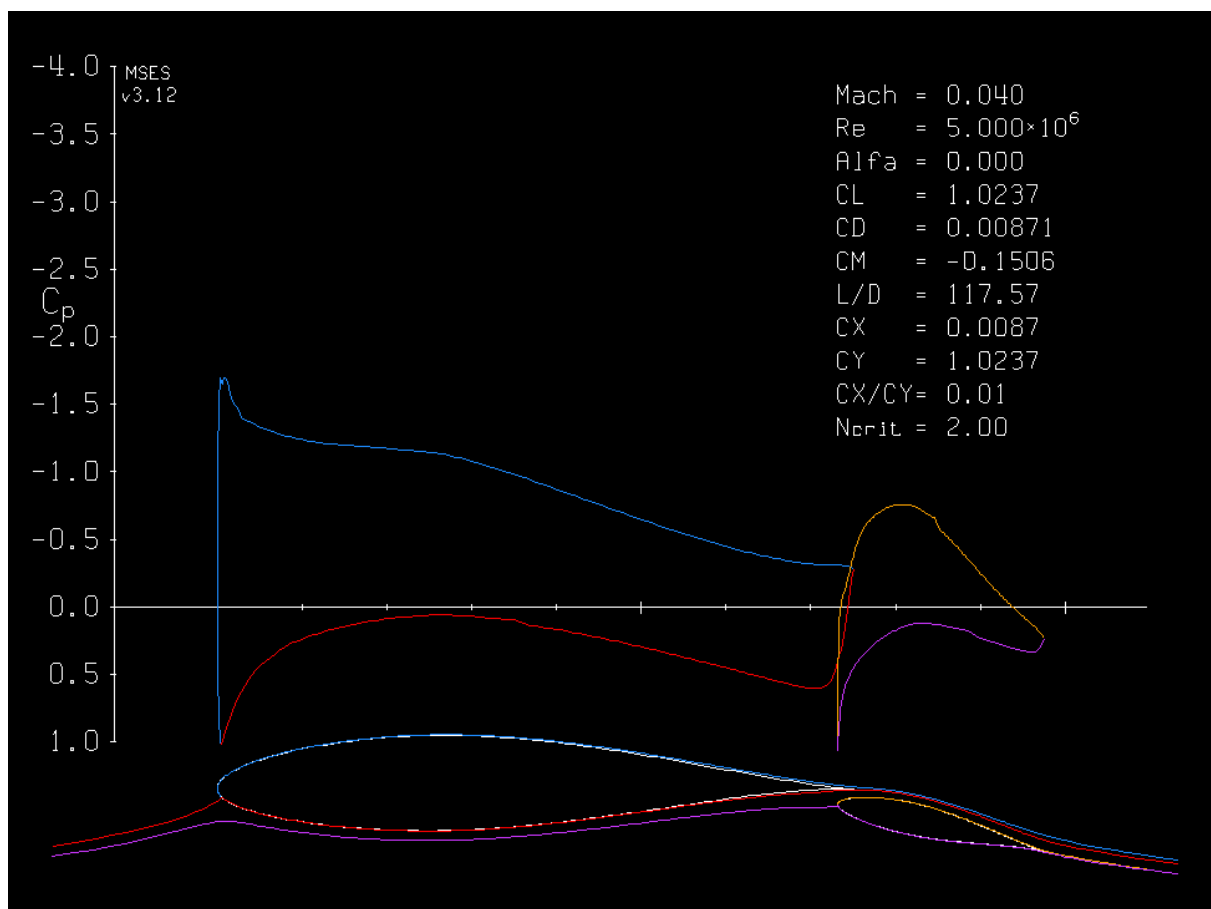


Figure 49: Sample pressure distribution taken from MSES ($Re = 5 \times 10^6$, $\alpha = 0^\circ$, Element 2 deflection = 12°).

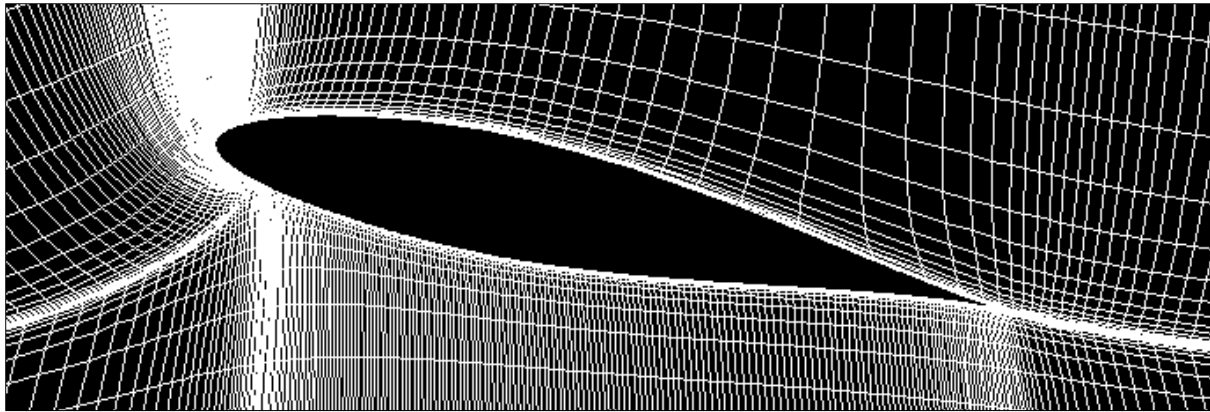


Figure 50: Intrinsic grid of a NACA 63-215 airfoil at $\alpha = 12^\circ$.

Appendix E-4 – MSES Lift and Drag Coefficients

The data taken from MSES and used in the MATLAB BEMT code is shown below in Table 8. The naming convention below is NACA63-215-X, where X is the deflection of the second element relative to the first element. Each configuration was tested between 0° - 20° angle of attack, with any change in this being due to limits of MSES.

Table 8: MSES data used in MATLAB BEMT code.

Airfoil	Angle of Attack	Coefficient of Lift	Coefficient of Drag
NACA63-215-0	0	0.20177	0.005454
NACA63-215-0	5	0.75654	0.008435
NACA63-215-0	10	1.29625	0.012707
NACA63-215-0	15	1.77715	0.023126
NACA63-215-1	0	0.26682	0.005474
NACA63-215-1	5	0.8199	0.008655
NACA63-215-1	10	1.35676	0.013106
NACA63-215-1	15	1.82946	0.02366
NACA63-215-2	0	0.33168	0.005559
NACA63-215-2	5	0.88436	0.008864
NACA63-215-2	10	1.41682	0.013522
NACA63-215-2	15	1.88077	0.024244
NACA63-215-3	0	0.39585	0.005793
NACA63-215-3	5	0.94781	0.009112
NACA63-215-3	10	1.4763	0.013965
NACA63-215-3	15	1.93082	0.024936
NACA63-215-4	0	0.45983	0.006058
NACA63-215-4	5	1.01083	0.009381
NACA63-215-4	10	1.53521	0.014413
NACA63-215-5	0	0.5241	0.006258
NACA63-215-5	5	1.07346	0.00967
NACA63-215-5	10	1.59333	0.014888
NACA63-215-6	0	0.5876	0.006585
NACA63-215-6	5	1.1358	0.009973
NACA63-215-6	10	1.65082	0.015387

NACA63-215-7	0	0.6503	0.007021
NACA63-215-7	5	1.19757	0.010283
NACA63-215-7	10	1.70764	0.015903
NACA63-215-8	0	0.71318	0.007358
NACA63-215-8	5	1.2587	0.010611
NACA63-215-8	10	1.7636	0.016442
NACA63-215-8	15	2.16228	0.029199
NACA63-215-9	0	0.77573	0.0077
NACA63-215-9	5	1.31928	0.010952
NACA63-215-9	10	1.81874	0.017014
NACA63-215-9	15	2.2052	0.03016
NACA63-215-10	0	0.83848	0.00793
NACA63-215-10	5	1.37803	0.011316
NACA63-215-10	10	1.87282	0.017597
NACA63-215-10	15	2.24638	0.031147
NACA63-215-10	20	2.37364	0.078686
NACA63-215-11	0	0.90076	0.008181
NACA63-215-11	5	1.43818	0.011677
NACA63-215-11	10	1.92594	0.018208
NACA63-215-11	15	2.28433	0.032206
NACA63-215-11	20	2.64235	0.002318
NACA63-215-12	0	0.96251	0.008433
NACA63-215-12	5	1.49664	0.012069
NACA63-215-12	10	1.97778	0.018843
NACA63-215-13	0	1.08426	0.008988
NACA63-215-13	5	1.60877	0.012851
NACA63-215-13	10	2.41519	0.001469
NACA63-215-14	0	1.02367	0.008707
NACA63-215-14	5	1.55426	0.012477
NACA63-215-14	10	2.02572	0.019374
NACA63-215-14	15	2.32724	0.037327
NACA63-215-15	0	1.1444	0.009265
NACA63-215-15	5	1.6653	0.013326
NACA63-215-15	10	2.12568	0.020875
NACA63-215-15	15	2.40416	0.036812
NACA63-215-16	0	1.20382	0.009568
NACA63-215-16	5	1.72288	0.013821
NACA63-215-16	10	2.17508	0.021623
NACA63-215-16	15	2.43364	0.038099
NACA63-215-17	0	1.26264	0.009889
NACA63-215-17	5	1.77746	0.014319
NACA63-215-17	15	2.34907	0.045018
NACA63-215-18	0	1.32445	0.010293
NACA63-215-18	5	1.83035	0.014855

NACA63-215-18	10	2.26534	0.023063
NACA63-215-19	0	1.37603	0.01067
NACA63-215-19	5	1.8813	0.015468
NACA63-215-19	10	2.10418	0.02017
NACA63-215-20	0	1.42972	0.011175
NACA63-215-20	5	1.93045	0.016115
NACA63-215-20	10	2.77395	0.06398
NACA63-215-21	0	1.47967	0.011799
NACA63-215-21	5	1.97431	0.016882
NACA63-215-21	10	2.59374	0.036154
NACA63-215-21	15	2.73143	0.045482
NACA63-215-22	0	1.52526	0.01253
NACA63-215-22	5	2.01857	0.017634
NACA63-215-22	15	2.61795	0.04598
NACA63-215-23	0	1.57503	0.013019
NACA63-215-23	5	2.0622	0.018319
NACA63-215-23	10	2.44588	0.028094
NACA63-215-23	15	2.29776	0.078346
NACA63-215-24	0	1.62148	0.013479
NACA63-215-24	5	2.10709	0.018991
NACA63-215-25	0	1.66828	0.014025
NACA63-215-25	5	2.14987	0.019678
NACA63-215-25	10	4.14996	0.061012
NACA63-215-26	0	1.71202	0.014593
NACA63-215-26	5	2.19031	0.020407
NACA63-215-26	15	2.44362	0.170413
NACA63-215-27	0	1.74769	0.015205
NACA63-215-27	5	2.227	0.021167
NACA63-215-28	0	1.77249	0.015961
NACA63-215-28	5	2.25694	0.021975
NACA63-215-28	15	2.79755	0.053512
NACA63-215-29	0	1.77466	0.01721
NACA63-215-29	5	2.27685	0.022907
NACA63-215-30	0	1.80038	0.018169
NACA63-215-30	5	2.27308	0.024238
NACA63-215-30	10	2.6126	0.03682
NACA63-215-30	15	1.1038	2.00E+10

Appendix E-5 - Pitch-Controlled Multi-Element Performance

Shown below is an estimate for the predicted power output curve if the power output stabilised at the rated wind speed by adjusting pitch, as discussed in Section 7.3.

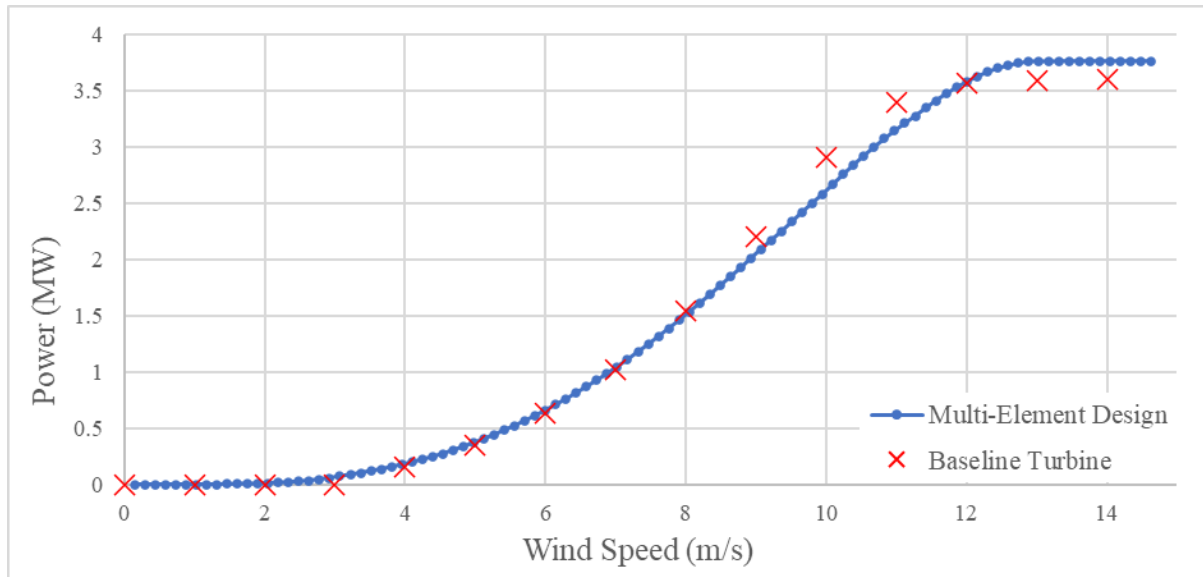


Figure 51: Estimate for the predicted power output curve of the multi-element design if the power output stabilised at the rated wind speed by adjusting pitch.

Appendix E-6 – Optimal Twist Distribution

In addition to outputting the graph for power coefficient against tip speed ratio, the MATLAB BEMT code outputs the optimal chord and twist distribution for the inputted turbine blade. The chord distribution is similar to the baseline and the twist distribution is similar to turbines of similar airfoil studies as highlighted in Figure 52 below, thus showing the feasibility of manufacturing a multi-element turbine blade (Taher *et al.* 2018).

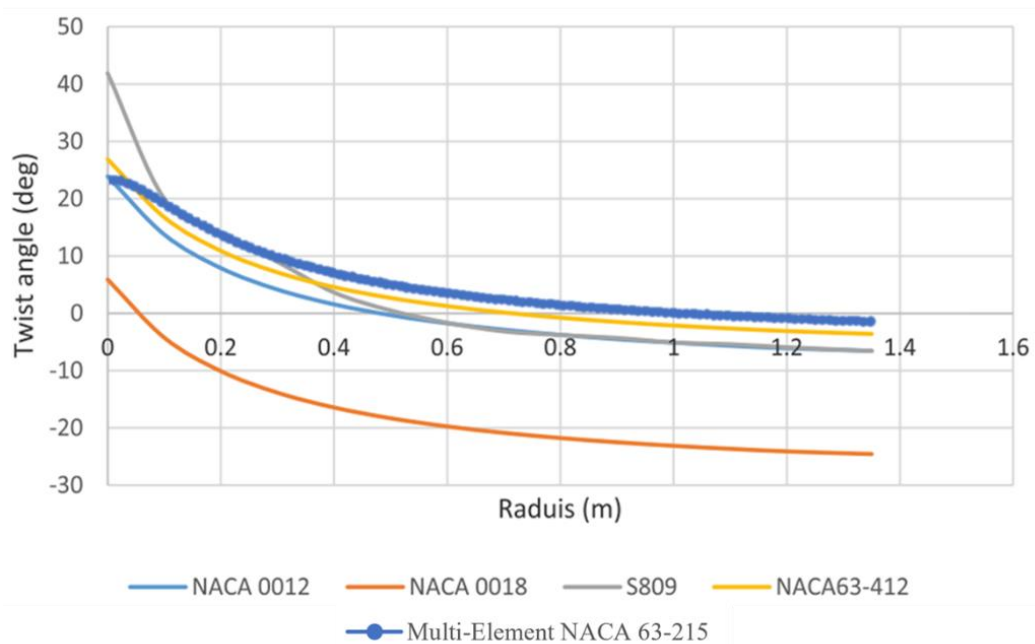


Figure 52: Twist angle for various turbines of different airfoils, including distribution of the proposed multi-element wind turbine design in this project to highlight manufacturing feasibility (Taher *et al.* 2018).

Appendix F – Further Equations for Prandtl's Approximation

Introducing the loss factor, as described in Section 4.5.1, into Equation 4 and 5 affects the following equations relating to induction factors and the power coefficient:

$$dT = 4a(1 - a)F\rho U_\infty^2 \pi r dr \quad (\text{Equation 4a})$$

$$dQ = 4a'(1 - a)F\rho U_\infty \Omega \pi r^3 dr \quad (\text{Equation 5a})$$

$$a = \frac{\sigma' C_L \cos \phi}{1 + 4F \sin^2 \phi} \quad (\text{Equation 21a})$$

$$a' = \frac{1}{\left[\frac{4 F \cos \phi}{\sigma' C_L} \right] - 1} \quad (\text{Equation 22a})$$

$$C_P = \frac{8}{\lambda^2} \int_0^\lambda F \lambda_r^3 a'(1 - a) \left[1 - \left(\frac{C_D}{C_L} \right) \cot \phi \right] d\lambda_r \quad (\text{Equation 23a})$$

Appendix G – MATLAB BEMT Code

The MATLAB BEMT code used within this project, originally developed by Salduskin (2013) and edited by Foale (2015), is shown below. The example input file is for the case with zero second element deflection of the multi-element HAWT.

```
clear
clc
%% Step 1 - Input Known Variables
% Length of Blades
Ri = 1.5; % Inner Radius - Axis of Rotation to Innermost Point on the Blade
Ro = 60; % Outer Radius - Axis of Rotation to Outermost Point on the Blade
r_step = 0.5; % Step Increments
r_vals = Ri:r_step:Ro;
r_vals_non_dim = r_vals/Ro;

% Load Hydrofoil Data from MSES
Hydrofoil=load('NACA63-215-0.dat');
AoA=Hydrofoil(:,1); % Angles of Attack
cl=Hydrofoil(:,2); % Coefficients of Lift
cd=Hydrofoil(:,3); % Coefficients of Drag

% Tip Speed Ratio Range & Increments
TSR_initial = 0.2;
TSR_final = 20;
TSR_step = 0.2;
TSR_vals = TSR_initial:TSR_step:TSR_final;

% Additional Design Parameters from Research Papers on Specific Airfoil Choice
Cl_design= 2.4; % Design Coefficient of Lift
TSR_design= 6.56; % Design Tip Speed Ratio
pitch= 6 ; % Design Angle of Attack (Degrees)
B = 3; % Number of Blades

% Specifications for Iteration Process
n = 100; % Maximum Number of Iterations
tol = 1e-3; % a and a_prime tolerance

% Iterate for TSR Values
for k = 1:length(TSR_vals)
    TSR(k) = TSR_vals(k);

% Create Empty Matrices for Each Blade Element
    a = zeros(length(r_vals), n);
    a_prime = zeros(length(r_vals), n);
    phi_deg = zeros(length(r_vals), n);
    alpha_deg = zeros(length(r_vals), n);
    phi = zeros(length(r_vals), n);
    alpha = zeros(length(r_vals), n);
    Ftot = zeros(length(r_vals), n);
    Ct = zeros(length(r_vals), n);
    Cl = zeros(length(r_vals), n);
```

```

Cd = zeros(length(r_vals), n);
Cl_2D=zeros(length(r_vals), n);
Cd_2D=zeros(length(r_vals), n);
cl_lin=zeros(length(r_vals), n);
Delta_cl=zeros(length(r_vals), n);

% Iterate for Each Blade Element
for i = 1:length(r_vals)
r(i) = r_vals(i);
lam_r(i) = TSR(k)*(r(i)/Ro);

% Chord Distributions
% Optimum Chord Distribution
c(i) = ((8/9)/(sqrt((1-
(1/3))^2+(TSR_design^2)*((r(i)/Ro)^2)*(1+(2)/(9*TSR_design^2*(r(i)/Ro)^2))^2)))*((
Ro*2*pi)/(TSR_design*Cl_design*B));
% Linear as in Burton et al. (2008)
% c(i) = Ro*(((8)/(9*0.8*TSR_design))*(2-
((TSR_design*(r(i)/Ro))/(0.8*TSR_design)))*((2*pi)/(Cl_design*TSR_design*B))));

% Optimum twist angle distribution
% theta(i) = (-pitch*pi/180) +atan((1-
(1/3))/(TSR_design*(r(i)/Ro)*(1+(2)/(9*(TSR_design)^2*(r(i)/Ro)^2))));
ThetaTwist(i)= atan((1-
(1/3))/(TSR_design*(r(i)/Ro)*(1+(2)/(9*(TSR_design)^2*(r(i)/Ro)^2))));
ThetaTwistDeg(i)= ThetaTwist(i)*180/pi;
ThetaPitch = (-pitch*pi/180);
%theta= thetaPitch+thetaTwist
theta(i) = ThetaPitch+ThetaTwist(i);
theta_deg(i)=(theta(i)*180)/pi;

% Rotor solidity
sigma_prime(i) = ( B*c(i) )/(2*pi*r(i));

%% Step 2 - Set Initial Induction Factors - a & a_prime Values
% Equation 4.21 Kacper Paper
% a_cor =0;
a_cor = real((1/4)* ( 2 + pi*lam_r(i)*sigma_prime(i) - sqrt(4-
4*pi*lam_r(i)*sigma_prime(i) +pi*lam_r(i)^2*sigma_prime(i) * (8*theta(i) +
pi*sigma_prime(i)) ) ));
a_prime_cor = 0;

% Iterate for Convergence of a & a_prime
for j = 1:n
a(i,j) = a_cor;
a_prime(i,j) = a_prime_cor;

%% Step 3 - Relative Flow Angle
% Equation 3.28
phi(i,j) = real(atan( (1 - a(i,j)) / ( ( 1+a_prime(i,j) ) * lam_r(i) ) ));
phi_deg(i,j) = (180/pi)*(phi(i,j));

```

```

%% Step 4 - Losses Associated with Tip & Root
% Equation 3.35
tmp1 = exp( -B/2 * (Ro - r(i)) / ( r(i) / Ro * sin( phi(i,j) ) ) );
Ftip(i,j) = real((2/pi) * acos( tmp1 ));

% Equation 3.36
tmp2 = exp( -B/2 * ( r(i) - Ri ) / ( r(i) * sin( phi(i,j) ) ) );
Froot(i,j) = real((2/pi) * acos( tmp2 ));
Ftot(i,j) = Ftip(i,j) * Froot(i,j);

%% Step 5 - AOA over Each Blade Element
%alpha(i,j) = phi(i,j) - theta(i);
alpha(i,j) = phi(i,j) - (ThetaTwist(i) + ThetaPitch);
alpha_deg(i,j) = (180/pi) * (alpha(i,j));

%% Step 6 - Determine Lift & Drag Coefficients
% Stall Delay Behaves Incorrectly for High c/r
c_over_r(i) = c(i) / r(i);
if c_over_r(i) > 0.7
    c_over_r(i) = 0.7;
end
Alpha_s = interp1(c_l, AoA, max(c_l), 'spline'); %Stall angle
c_l_s = interp1(AoA, c_l, Alpha_s, 'spline'); %Cl at stall

% Determine Linear Portion of Cl vs Alpha curve (for stall delay)
x1 = -4;
x2 = 4;
y1 = interp1(AoA, c_l, x1, 'spline');
y2 = interp1(AoA, c_l, x2, 'spline');
m = (y2 - y1) / (x2 - x1); %Slope

% Determine 2D Coefficients
Cl_2D(i,j) = interp1(AoA, c_l, alpha_deg(i,j), 'spline');
Cd_2D(i,j) = interp1(AoA, c_d, alpha_deg(i,j), 'spline');
c_l_lin(i,j) = m * (alpha_deg(i,j) - x1) + y1;
Delta_cl(i,j) = c_l_lin(i,j) - Cl_2D(i,j); % Difference between viscous and inviscid
lift coefficients

% Lift & Drag Extrapolation (Viterna method)
AR = 10; % Aspect ratio of blade
Cd_max = 1.11 + 0.018 * AR; % Cdmax = Coefficient B1 to be used in the viterna method
%Maximum dragcoefficient %Alpha=90 degrees
Cl_max = max(c_l) + 3 * ((c_over_r(i))^2) * ((m * (Alpha_s - x1) + y1) - Cl(5)); % Maximum lift
coefficient
Cd_s = interp1(AoA, c_d, Alpha_s, 'spline'); % Drag coefficient at stall
A1 = Cd_max / 2; % Coefficient A1 to be used in the viterna method
A2 = (c_l_s -
Cd_max * sind(Alpha_s) * cosd(Alpha_s)) * ((sind(Alpha_s)) / ((cosd(Alpha_s))^2));

```

```

% Coefficient A2 to be used in the Viterna Method
B2=(Cd_s-Cd_max*((sind(Alpha_s))^2)/cosd(Alpha_s); %Coefficient B2 to be used in
the viterna method.
x11=90; y11=0; x21=180-Alpha_s; y21=-0.7*Cl_max; m1=(y21-y11)/(x21-x11);
x12=180; y12=0; m2=(y21-y12)/(x21-x12);
x13=-10; y13=-0.7*Cl_max;
x23=-180+Alpha_s; y23=0.49*Cl_max; m3=(y23-y13)/(x23-x13);
x14=-180; y14=0; m4=(y23-y14)/(x23-x14);

% Convert 2D Data to 3D (Stall delay)
if alpha_deg(i,j)>=-8 && alpha_deg(i,j)<=14

% 3D Data (Stall delay) *Snel Method
Cl(i,j)=Cl_2D(i,j)+3*((c_over_r(i))^2)*Delta_cl(i,j); % Emperical method for stall
delay (Snel et al.)
Cd(i,j)=Cd_2D(i,j); %Drag is assumed to be equal to 2D data

% Between 13 degrees & 90
elseif alpha_deg(i,j)>14 && alpha_deg(i,j)<=90

% Viterna Method Applied on Angles Outside the Attached Flow Region.
Cl(i,j)=Cd_max*sind(alpha_deg(i,j))*cosd(alpha_deg(i,j))+A2*(((cosd(alpha_deg(i,j))
)^2))/((sind(alpha_deg(i,j)))));
Cd(i,j)=Cd_max*(sind(alpha_deg(i,j))^2)+B2*cosd(alpha_deg(i,j));

% Between 90 & 180 - Stall Angle
elseif alpha_deg(i,j)>90 && alpha_deg(i,j)<180-Alpha_s
    Cl(i,j)=m1*(alpha_deg(i,j)-x11)+y11;
    Cd(i,j)=Cd_max*(sind(alpha_deg(i,j)))^2+B2*cosd(alpha_deg(i,j));

% Between 180 - Stall & 180
elseif alpha_deg(i,j)>=180-Alpha_s && alpha_deg(i,j)<=180
    Cl(i,j)=m2*(alpha_deg(i,j)-x21)+y21;
    Cd(i,j)=Cd_max*(sind(alpha_deg(i,j)))^2+B2*cosd(alpha_deg(i,j));

% Between -8 & -180 plus Stall
elseif alpha_deg(i,j)<-8 && alpha_deg(i,j)>=-180+Alpha_s
    Cl(i,j)=m3*(alpha_deg(i,j)-x13)+y13;
    Cd(i,j)=Cd_max*(sind(alpha_deg(i,j)))^2+B2*cosd(alpha_deg(i,j));

% between
elseif alpha_deg(i,j)<-180+Alpha_s && alpha_deg(i,j)>=-180
    Cl(i,j)=m4*(alpha_deg(i,j)-x23)+y23;
    Cd(i,j)=Cd_max*(sind(alpha_deg(i,j)))^2+B2*cosd(alpha_deg(i,j));
end

%% Step 7 - Calculate coefficient of thrust
% Equation 4.28 Kacper Paper
Ct(i,j) = ( sigma_prime(i) * (1-a(i,j))^2 * ( Cl(i,j) * cos(phi(i,j)) + Cd(i,j) *
sin(phi(i,j))) ) / (sin(phi(i,j))^2) ;

```



```

% Update new induction factors
% From Paper by (R. Lanza fame, M. Messina, 2007)
if Ct(i,j) < 0.96*Ftot(i,j)
a_cor1 = ( 1 + ( 4 * Ftot(i,j) * (sin(phi(i,j)))^2 / ( sigma_prime(i) * Cl(i,j) *
cos(phi(i,j))+Cd(i,j)*sin(phi(i,j)) ) ) ) ^(-1);
a_cor = 0.2*a_cor1 + 0.8*a(i,j);
a_prime_cor = (1/2) * ( sqrt(1 +(4/lam_r(i)^2) * a_cor * (1 - a_cor)) - 1 );

% Sam Whelans Paper Eq 4.24 Glauerts Correction Near the Tip
% Buhl Correction From Paper by (R. Lanza fame, M. Messina, 2007) Eq 7
elseif Ct(i,j) > 0.96*Ftot(i,j)
a_cor1 = ( 18*Ftot(i,j) - 20 - 3 * sqrt( Ct(i,j)*(50-36*Ftot(i,j)) +
12*Ftot(i,j)*(3*Ftot(i,j) - 4) ) ) / (36*Ftot(i,j) - 50); % from paper by
Lanza fame, Messina
% a_cor = 0.2*a_cor1 + 0.8*a(i,j);
a_cor = 0.5*a_cor1 + 0.5*a(i,j);

% Eq 7
a_prime_cor = (1/2) * ( sqrt(1 +(4/lam_r(i)^2) * a_cor * (1 - a_cor)) - 1 ); %
from paper by Lanza fame, Messina
end

%% Step 8 - Assess Tolerance of New Induction Factors to Previous
% Induction Factors
if (abs((a(i,j) - a_cor) / a_cor) <= tol)
break;
end
end

%% Step 9 - Calculate Blade Element Coefficient of Power
Cp(k,i) = ( Ftot(i,j) * sin(phi(i,j))^2 * ( cos(phi(i,j)) -
(lam_r(i)*sin(phi(i,j))) ) * (sin(phi(i,j)) +(lam_r(i)*cos(phi(i,j))) ) * (1-
(Cd(i,j)/Cl(i,j))*cot(phi(i,j))) * lam_r(i)^2 );
end

% Solve for Rotor Coefficient of Power at Specified TSR
Cp_sum = real(8 / (TSR(k) * length(r_vals) ) ) * sum(Cp(k,:));
Cp_range(k) = real(Cp_sum);
CPReal= Cp_range * 0.87;
end

%% Step 10 - Graph results
figure;
plot(TSR_vals, CPReal, 'bo-');
xlabel('TSR');
ylabel('Cp');
xlim([0 10]);
ylim([0 0.6]);
grid on

```

```
figure;  
plot(r_vals_non_dim, c, 'bo-');  
xlabel('r/Ro');  
ylabel('Chord');  
xlim([0 1]);  
ylim([0 2]);  
grid on  
  
figure;  
plot(r_vals_non_dim, theta_deg, 'bo-');  
xlabel('r/Ro');  
ylabel('Twist');  
xlim([0 1]);  
ylim([0 40]);  
grid on
```